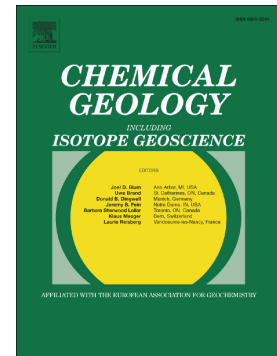


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PII: S0009-2541(25)00215-3

DOI: <https://doi.org/10.1016/j.chemgeo.2025.122825>

Reference: CHEMGE 122825

To appear in: *Chemical Geology*

Received date: 26 July 2024

Revised date: 27 April 2025

Accepted date: 2 May 2025

Please cite this article as: Y. Wang, M. Yang, X. Han, et al., Trace element and sulfur isotope study of sulfide chimneys from the mafic-hosted Daxi Vent Field: Implications on the genesis of seafloor massive sulfide deposit at a segment end, *Chemical Geology* (2024), <https://doi.org/10.1016/j.chemgeo.2025.122825>

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Trace element and sulfur isotope study of sulfide chimneys from the mafic-hosted Daxi Vent Field: implications on the genesis of seafloor massive sulfide deposit at a segment end

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Abstract

The mafic-hosted Daxi Vent Field (DVF) on the Carlsberg Ridge hosts an actively forming massive sulfide deposit on the neo-volcanic ridge at a non-transform offset (NTO). Here, we present mineralogy, geochemistry and in situ sulfur isotope analysis of the sulfide chimneys sample from the DVF, which can be classified into Fe-rich and Zn-rich types based on the major sulfide minerals. Our results show that the Fe-rich chimneys, primarily consisting of pyrrhotite and isocubanite, exhibit enrichment in Co, Se, and Ni, with heavy $\delta^{34}\text{S}$ in pyrite (up to 17.38 ‰). The highly saline fluids resulting from phase separation processes lead to the significant Co enrichment. The precipitation of pyrite in a relatively closed environment intensified by the Rayleigh fractionation, leading to significant heavy S isotopic composition. On the other hand, the Zn-rich chimneys are predominantly sphalerite, and enriched in Sn, Ag, As, Ga, Ge, and Cd. The fluid-seawater mixing processes contribute to the elevated Sn levels and the wide range of S isotopic compositions ($\delta^{34}\text{S} = 1.47\sim 8.34\text{‰}$) in sphalerite. The unusual Co and Sn enrichment as well as unique S isotopic composition in the DVF are linked to the thin ocean crust at NTOs, where the thin ocean crust may favor interaction of fluids circulating in the subsurface with ultramafic rocks, despite the absence of ultramafic lithologies on the seafloor. This study demonstrates the exploration potential for polymetallic resources in the NTO along slow-spreading mid-ocean ridges.

Keywords: Seafloor massive sulfide deposit; Trace element; Sulfur isotope; Non-transform offset

1. Introduction

The slow- and ultraslow-spreading mid-ocean ridges with full spreading rates of less than 55 mm/yr have a combined length of approximately 35,000 km, accounting for nearly 60% of the total length of global mid-ocean ridges (Beaulieu et al., 2015). In these settings, the combination between discontinuous magma supply and tectonic processes leads to a highly heterogeneous oceanic lithosphere, uplifting or exposing lower-crustal and upper-mantle rocks (Früh-Green et al., 2022). They are considered to have a great potential for seafloor polymetallic sulfide resource exploration due to sustained fluid circulation along deep-penetrating and continuously reactivated detachment fault systems (e.g., Hannington et al., 2011; German et al., 2016; Anderson, 2018). Previous studies have indicated that the mineralization of hydrothermal systems distributed on slow-spreading ridges is predominantly influenced by various factors such as local magmatic activity, tectonic processes, and the characteristics of surrounding rocks (Hannington et al., 2005). Therefore, the seafloor massive sulfide (SMS) deposits associated with the hydrothermal systems at mid-ocean ridges can be classified as magmatic- or tectonic-hosted ones (German et al., 2016). For instance, at the center of a second-order segment, seafloor hydrothermal circulation is driven by magma chambers and/or lenses at depth (Jian et al., 2017), and the resulting high-temperature fluids are transported upward to the seafloor. The subsurface mineralization occurs along channels, such as high-angle normal faults within the oceanic crust (e.g., Nakamura et al., 2012; German et al., 2016). These mafic-hosted SMS deposits are typically characterized by Zn and Fe enrichment (Fouquet et al., 2010; Petersen et al., 2018; Cai et al., 2024) and have been defined as magmatic-controlled deposits by German et al. (2016). Conversely, at the segment-ends, where the oceanic crust is thinner and tectonic activity is more intense, the development of detachment faults and serpentinization in the upper mantle influences the circulation and physicochemical characteristics of the hydrothermal fluids (e.g., Escartín et al., 2008; Nakamura et al., 2009; Wang et al., 2014; German et al., 2016; Tao et al., 2020; Früh-

Green et al., 2022). Hydrothermal fluids interacting with ultramafic rocks generate H_2 and CH_4 , creating highly reducing conditions that favor the formation of reduced sulfide assemblages, including pyrrhotite, isocubanite, and Fe-rich sphalerite (Charlou et al., 2002, 2010; Nakamura et al., 2009). The so-called tectonic-controlled, mostly ultramafic-hosted, SMS deposits formed in this geological setting are normally more enriched in Cu, Zn, Co, Ni, and Au, primarily due to the higher abundance of these metals in ultramafic rocks (Fouquet et al., 2010; German et al., 2016; Patten et al., 2021; Yang et al., 2024).

In 2015, the Daxi Vent Field (DVF) was discovered within massive sulfide accumulations along the slow-spreading Carlsberg Ridge at 6°48'N, 60°10'E (Wang et al., 2021). Its distinction lies in its location on a non-transform offset (NTO) and being mafic-hosted. To date, only nine hydrothermal fields with a similar geological setting have been reported out of nearly 700 hydrothermal sites in the world's oceans (Wang et al., 2021). The sulfide chimneys in the DVF are significantly enriched in Zn, Co and Sn but deficient in Cu and Au (Wang et al., 2021), which are different from the metal enrichment characteristics of the typical mafic- and ultramafic-hosted SMS deposits along slow- and ultraslow-spreading ridges (Fig. 1A; Fouquet et al., 2010; Patten et al., 2021). Further study of the hydrothermal fluid chemistry from the DVF indicate significant enrichment in H_2 (7.07 mM), CH_4 (0.88 mM), and Sn (up to 0.018 ppm), suggesting the participation of upper mantle peridotites and/or lower crustal gabbro in the seafloor hydrothermal circulation processes (Wu et al., 2022). Recent research at the Aurora Vent Field on the ultraslow-spreading Gakkel Ridge (Arctic Ocean) has corroborated that fluid chemistry in a magmatic-setting may be affected by deep ultramafic lithologies beneath unusually thin oceanic crust during fluid circulation (German et al., 2022). The geological setting that thin layers of basaltic seafloor overlying deeper crustal and mantle rocks may be a recurring feature along the slow- and ultraslow-spreading ridges (Albers et al., 2025). Therefore, it is plausible that hydrothermal mineralization formed at the termination of a segment, along with its

controlling mechanism, diverges from the conventional comprehension of ore-forming process at slow-spreading ridges.

In this study, we conduct detailed mineralogical, mineral chemistry, and in situ sulfur isotope analyses on samples of black smokers collected from the DVF in order to determine the physicochemical conditions of sulfide chimney formation and metal enrichment. We discuss the genetic implications of mineralization at the NTOs and suggest that this special geologic setting contributed to the formation of Zn-rich chimneys and Fe-rich chimneys, as well as unique Co and Sn enrichment in the DVF.

2. Previous work

2.1. Geological setting

The Carlsberg Ridge (CR) is a slow-spreading ridge with a full spreading rate of 22–32 mm/yr. It is adjusted by lateral displacement and shear and sub-divided by right-stepping NTOs (Murton & Rona, 2015). Three second-order segments are separated by two NTOs along the CR between 6°N and 7°N (Fig. 1B; Wang et al., 2021). The southernmost segment (segment-1) features a 40 km long, NW-SE oriented axial volcanic ridge (AVR) situated within the approximately 8 km wide central valley at water depths ranging from 2900 to 3500 m. Within this segment's neo-volcanic ridge, two mafic-hosted hydrothermal systems, Wocan-1 and Wocan-2, are present (Wang et al., 2017). The northern segment (segment-2) extends for 30 km and features a relatively narrow axial valley (~7 km wide) with greater depths ranging from 3350 to 3750 m, along with a small AVR measuring 300 m in height and 24 km in length. The most northern segment (segment-3) covers a distance of 60 km and includes a rift valley approximately 8 km wide, situated at water depths ranging from 2250 to 3750 m, housing a significant AVR. (Wang et al., 2021).

The DVF (6°48'N/60°10'E, water depth ~3450 m) is located on a neo-volcanic ridge between segment-2 and segment-3 (Fig. 1B). This NTO is oriented near southeast

by east, displaces two second-order segments by ~13 km and developing a neo-volcanic ridge in its center (Wang et al., 2021). The Daxi volcanic ridge measures approximately 5.6 km in length, 2.4 km in width, and 500 m in height, featuring a 130 m deep V-shaped rift valley (Fig. 1C). Multiple flat-topped volcanoes, reaching heights of up to 350 m and diameters of 1300 m, are distributed on either side of the neo-volcanic ridge. This segment can be subdivided into second-order ridges, with consistent mantle source composition attributed to the depleted peridotite mantle (Hu et al., 2022).

During the DY38I cruise in 2017 (Wang et al., 2021), the manned submersible *Jiaolong* was used to carry out near-bottom optical observations, high-resolution bathymetric surveys, and rock and fluid sampling. In situ observations revealed that the area is dominated by a central mound (~180 m in diameter; water depth ~3500 m) and a northeast mound (diameter of ~60 m; water depth of ~3400 m), which are controlled by two high-angle NW-trending normal faults that extend parallel to the strike of the neo-volcanic ridge (Wang et al., 2021). Eight active hydrothermal vent clusters are distributed randomly on the central mound (Fig. 2A). A large number of active and inactive sulfide chimneys and their collapse deposits with different morphologies (e.g., pipe-like chimney and beehive-like chimney) occur nearby (Figs. 2B–E). Field observations and geological sampling revealed that the rocks in the neo-volcanic ridge and adjacent areas of the hydrothermal field are mainly composed of fresh basaltic lavas and brecciated pillows, with no upper mantle peridotites and a few thin layers of pelagic sediments covering their surfaces, implying recent volcanic activity (Wang et al., 2021).

The relationship between ridge tectonic segmentation, axial depth, ridge morphology and underlying magma system is well-established (Murton & Rona, 2015; Carbotte et al., 2015). The mantle Bouguer anomaly (MBA; Fig. S1B) highlights crustal thickness variations (Guo et al., 2023): segment-1 (–19.3 mGal) and segment-3 (–15.9 mGal) have lower values than segment-2 (–3.85 mGal). At the DVF near NTO-2, with elevations >500 m (Fig. S1C), the MBA averages 7.77 mGal (Fig. S1B). This indicates

that the surface eruptions in the DVF is currently primarily influenced by magma supply, while the deep magma supply is still contributing to the formation of the lower oceanic crust (Han et al., 2012). The underlying mantle of the ridge in the DVF undergo 18% partial melting (Fig. S1A) and higher pressure of melting (8.32 kbar), which aligns with its thinner oceanic crust and the tendency of melts to crystallize at greater depths within the NTO-2 (Hu et al., 2025, accepted).

2.2. *Hydrothermal fluid characteristics*

Hydrothermal fluids at Buddhas Hand and A1 chimneys reach temperatures of 273 and 272°C, respectively (Wu et al., 2022). The DVF fluids are characterized by a depletion of B (1.28 mM; Wu et al., 2022), which may indicate that the existence of serpentinization somewhere along the path of seafloor circulation (Seyfried et al, 1984; Boschi et al, 2008; Schmidt et al, 2011). The fluids show varying metal content relative to each other with high Fe, Mn, Cu, and Co content at Buddhas Hand while Zn, Ga, Cd, Ag, and Sn co-vary with each other and are strongly enriched at A1 (Wu et al., 2022). This strong enrichment in most of the transition and heavy metals in the fluid has been observed in both mafic and ultramafic hydrothermal systems and indicates phase separation into vapor and brine phases (Douville et al., 2002; Webber et al., 2015; Schmidt et al., 2017). Additionally, Buddha's Hands fluid is highly Cl-enriched (928 mM), while the A1 fluid is Cl-depleted (303 mM) indicating phase separation (Wu et al., 2022). Another striking feature of their vent fluid chemistry is the high CH₄ (0.88 mM), H₂ (7.07 mM) concentration. This is similar to that of the Nibelungen (Melchert et al., 2008) and Kairei (Gallant & Von Damm, 2006; Kumagai et al., 2008) vent fields, which are located in segment-end settings along the southern Mid-Atlantic Ridge and Central Indian Ridge, respectively.

2.3. *Mineralization characteristics*

Sulfide chimneys located on the central mound in the DVF are distinguished

between Zn-rich and Fe-rich chimneys (Wang et al., 2021). The Zn-rich chimneys, which often have a beehive-like structure on the lower section of the chimney (Fig. 2D), are dominated by sphalerite (65%) and pyrrhotite (20%) with minor chalcopyrite (10%) and isocubanite (5%) (Table 1). These chimneys are characterized by enrichment of Zn, Ag, Sn, Cd, and Pb, with average contents ($n = 27$) of 45 wt.%, 278 ppm, 1309 ppm, 1163 ppm, and 1669 ppm, respectively. The Fe-rich chimneys mainly consist of branched structures topped by multiple pipe-like spires (Fig. 2C), with the primary mineral assemblage being pyrrhotite (70%) and pyrite (20%), along with minor quantities of chalcopyrite (6%) and sphalerite (4%) (Table 1). These chimneys are characterized by relative enrichment in Fe (avg. 43.8 wt.%; $n = 12$), Cu (avg. 4.84 wt.%), Au (avg. 2.60 ppm), Co (avg. 3452 ppm), and As (avg. 106 ppm) (Wang et al., 2021).

3. Samples and Analytical Methods

Samples were collected from 12 stations on the Central Mound in the DVF (see Fig. 2A and Table 1 for sampling locations) by both submersible and TV-grab methods. The Zn-rich chimneys are porous in texture, with a yellowish-brown surface due to iron oxidation (Figs. S2A–B). Unlike the highly porous beehive structure of the Zn-rich chimneys (Fig. S2C), the Fe-rich chimneys are less porous and commonly contain one primary fluid channel (Fig. S2D). Thirty-three polished thin sections from all chimney samples were examined using a reflected and transmitted light polarizing microscope at the Second Institute of Oceanography, Ministry of Natural Resources (SIOMNR) in Hangzhou, China, and were subsequently analyzed via X-ray diffraction (XRD).

3.1. Mineral chemistry analyses

Electron microprobe analyses (EPMA) of individual sulfide minerals were carried out using a JEOL JXA-8100 Superprobe at the SIOMNR, operated with an acceleration voltage of 20 kV, a beam current of 1 nA, and a beam diameter of 5 μm . A variety of natural and synthetic minerals (USA SPI Supplies) have been used as standards. These include chalcopyrite (CuFeS_2), sphalerite (ZnS), gallium arsenide (GaAs), as well as

the pure metals silver (Ag), cobalt (Co), nickel (Ni), cadmium (Cd), antimony (Sb), and selenium (Se). The results of the individual EPMA analyses are presented in Table S1. The trace element compositions of the sulfide minerals were analyzed via laser ablation-inductively coupled plasma-mass spectrometry (LA-ICP-MS) at GEOMAR, Helmholtz Center for Ocean Research Kiel, Germany and the Institute of Geochemistry, Chinese Academy of Sciences, China. The LA-ICP-MS analyses at GEOMAR were performed using a 193 nm Excimer laser ablation system (Coherent, GeoLasPro) coupled to a double-focusing, high-resolution magnetic sector mass spectrometer (Nu Instruments, AttoM). Ablation was performed under He carrier gas, additionally Ar carrier gas was mixed in prior to the plasma torch (Grant et al., 2018). The ablation beam diameter was 44 μm , depending on grain size, with a pulse rate of 5 Hz and energy fluence of 2 J cm^{-2} . The total analysis time for each spot was 70 s, with 40 s of background measurement and 30 s of sample ablation. MASS-1 (USGS), trans-1, and BHVO-2G (USGS) were used as reference materials. The trace element concentrations were also determined by LA-ICP-MS at the Institute of Geochemistry, Chinese Academy of Sciences, with an NWR-213 (New Wave Research) laser ablation system coupled to an Agilent 7700x ICP-MS instrument. He (350 ml/min) was applied as the carrier gas, and Ar (900 ml/min) was used as the make-up gas. All analyses were run with a 26 μm pit size, 5 Hz pulse frequency and 3 J cm^{-2} fluence. Each analysis incorporated a background acquisition of approximately 30 s (gas blank) followed by 60 s of data acquisition. The concentration was calculated using multiple external standards and sum normalization method (Liu et al., 2008). The external standards STDGL3, GSD-1G, and natural pure pyrite, and MASS-1 (USGS synthetic reference material) were analyzed as a control to check the data quality. The complete datasets, including the detection limits and reference materials for all elements analyzed via LA-ICP-MS, are presented in Table S2, S3, and S5, respectively.

3.2. Sulfur isotope analyses

In situ sulfur isotope analyses of 166 spots on sulfide minerals were performed using the sensitive high resolution ion microprobe-stable isotope (SHRIMP-SI) at the Research School of Earth Sciences, Australian, as described by Ireland et al. (2014). A beam of primary Cs^+ ions was focused to 6 nA with an acceleration voltage of 15 kV. The secondary ion mass spectrometry analyses of the sphalerite, pyrite, chalcopyrite, and pyrrhotite were conducted using a beam depth of $\sim 1 \mu\text{m}$ and a spot size of $\sim 35 \mu\text{m}$. The sulfur isotope data for the sphalerite, pyrite, pyrrhotite, and chalcopyrite were normalized to the accepted standard Balmat sphalerite, Ruttan pyrite, Anderson pyrrhotite, and Norilsk chalcopyrite, which have $\delta^{34}\text{S}_{\text{VCDT}}$ values of 14.3‰, 1.2‰, 1.4‰, and 8.0‰, respectively (Crowe & Vaughan, 1996). The sample measurements were bracketed by at least one standard measurement every three to four analyses to ensure precision and to account for instrumental drift. The spot-to-spot reproducibility (external precision) was typically better than $\pm 1.28\text{‰}$ (2σ) for sphalerite, $\pm 0.52\text{‰}$ (2σ) for pyrite, $\pm 0.27\text{‰}$ (2σ) for pyrrhotite, and $\pm 0.70\text{‰}$ (2σ) for chalcopyrite. The complete sulfur isotope dataset for the sulfide minerals and standards is presented in Table S4.

4. Results

4.1. Mineralogy

Zn-rich chimneys: The Zn-rich chimneys have the following paragenetic sequence: pyrrhotite + sphalerite-I $\rightarrow$ chalcopyrite + isocubanite + sphalerite-II $\rightarrow$ sphalerite-III + pyrite/marcasite $\rightarrow$ goethite + iron oxyhydroxides + amorphous silica (Fig. 3). The mineral assemblage of the Zn-rich chimneys is controlled by fluid channels and exhibits distinct mineral zonation. The main sulfide minerals are sphalerite (from early to late generation: Sph-I, Sph-II, Sph-III), isocubanite, pyrrhotite, and chalcopyrite, and the gangue minerals are predominantly opal. The early precipitated high-temperature subhedral pyrrhotite within the fluid channels is often replaced by medium- to high-temperature sphalerite (Sph-I)-chalcopyrite. The rhythmic growth layer of the

chalcopyrite is shown in Fig. 4A. The Sph-I in the interior is porous and paragenetic with small isocubanite particles, while the Sph-II at the edge is homogeneous and idiomorphic, and the two phases of sphalerite are separated by isocubanite (Fig. 4B). Opal acts as a late cementing mineral of the earlier formed sulfide minerals such as sphalerite (Fig. 4C). The areas close to the fluid channels of the chimney are dominated by idiomorphic and subhedral sphalerite (Sph-II) and are often paragenetic with high-temperature minerals such as lath-shaped pyrrhotite (Fig. 4D), isocubanite, and chalcopyrite (Fig. 4E). The outer layer of the chimney primarily consists of colloform sphalerite (Sph-III) with a zonal structure and occasional chalcopyrite disease structures (Fig. 4F).

Fe-rich chimneys: The Fe-rich chimneys have the following paragenetic sequence: pyrrhotite + sphalerite + chalcopyrite + isocubanite → pyrite + covellite → iron oxyhydroxides + amorphous silica (Fig. 3). The main sulfide minerals are pyrrhotite, isocubanite, and chalcopyrite, and the minor mineral are sphalerite, pyrite, and iron hydroxide. The lath-shaped pyrrhotite, around the fluid channels, replaced by chalcopyrite (Fig. 4G). In the intermediate part of the chimney, the pyrrhotite is often paragenetic with the isocubanite and chalcopyrite (Fig. 4H). Some pyrrhotite is paragenetic with the subhedral sphalerite and the minor pyrite and covellite (Fig. 4I). Notably, the oxidation of the pyrrhotite was prevalent with visible pores on the surface (Figs. 4H-I). Furthermore, the later allotriomorphic pyrite formed along the cracks in the chalcopyrite (Fig. 4J). The subhedral pyrite precipitated in the fractures of breccias (Fig. 4K). The outer layer of the chimney was oxidized, and the pyrrhotite was filled with opal forming a skeleton texture. Subsequently, a large amount of iron hydroxide was formed with some pyrrhotite completely oxidized (Fig. 4L).

4.2. Mineral chemistry

Sphalerite (ZnS): Sphalerite is the most abundant mineral in the sulfide chimney samples. The Fe wt.% and FeS mole% content of the different sphalerite phases varies

in the Zn-rich chimneys (Fig. 5; Table S1). The Sph-I has a Fe concentration of 9.23–17.92 wt.% (median = 15.44 wt.%, $n = 10$), with FeS mole% of 16.14–30.75 mol% (median = 26.60 mol%). The Fe concentration in Sph-II is 6.41–17.39 wt.% (median = 8.50 wt.%, $n = 24$) and the FeS mole% is 11.05–29.22 mol% (median = 14.87 mol%). The Sph-III has a Fe concentration of 1.18–7.00 wt.% (median = 4.06 wt.%, $n = 41$), with FeS mole% of 2.07 to 12.36 mol% (median = 7.15 mol%). In particular, the late colloform sphalerite (Sph-III) has a zoned structure and covariate Fe, Zn, and Sn content trends as shown by the EPMA element maps (Fig. 6). Compared to that from the Zn-rich chimneys, the sphalerite from the Fe-rich chimneys has a narrower range of Fe contents (5.11–18.3 wt.%), with a higher median of 9.38 wt.% ($n = 27$) (Table S1). The Fe-rich/medium sphalerite ($n = 26$) from Fe-rich chimneys (Fig. 5A) with FeS concentrations of 9.01 to 31.0 mol%, with an average of 18.4 mol% (Table S1).

We analyzed 71 sphalerite grains from Zn-rich chimney ($n = 51$) and Fe-rich chimney ($n = 20$) samples, using LA-ICP-MS to identify the distribution trends of the trace elements. The LA-ICP-MS line scan analyses of the sphalerite revealed a covariation trend for Mn-Ag-Fe-Pb and Co-Cu-Sn-Cd (Fig. S3A). Compared to the pyrrhotite and isocubanite, the median values of the Sn, Ag, Cd, and Pb contents of the sphalerite are 1 to 3 orders of magnitude higher (Fig. 7A). However, different generations of sphalerite have significant differences in trace element compositions (Fig. 7B). Sph-I has higher Fe, Cu, Cd, Sn, and Sb contents compared to Sph-II and Sph-III, but lower Ag contents. The average Co content of the Sph-II is the highest, with a maximum value of up to 1486 ppm. With high Ag and Pb but low Fe and Co concentrations, Sph-III is characterized by the enrichment in Cu, Ga, Sn, and Tl in the core and Ge in the margin (Table S2). In the Fe-rich chimneys, the sphalerite ($n = 20$) is relatively enriched in Co and Sn and deficient in Ag and Pb (Fig. 7B). The median Cu, Cd, Sn, and Pb contents are >1000 ppm, with values of 2835 ppm, 1935 ppm, 1414 ppm, and 1211 ppm, respectively (Table S2).

Pyrrhotite ($Fe_{1-x}S$): The EPMA results show that the Fe contents of the pyrrhotite from the Zn-rich chimneys are 59.1–60.8 wt.% ($n = 18$). The Fe contents of the

pyrrhotite from the Fe-rich chimneys are 59.0–60.2 wt.% ($n = 17$), which is close to that of the Zn-rich chimneys (Table 2). The LA-ICP-MS analyses of the pyrrhotite ($n = 15$) from the Zn-rich chimneys revealed a high median Co content (median = 4806 ppm) and a low Cu content (median = 935 ppm). The As, Mo, Ag, and Sn contents exhibit increasing trends in several minerals, which may be associated with the sphalerite inclusions in the pyrrhotite (Fig. S3C). Compared to that from the Zn-rich chimneys, the pyrrhotite ($n = 14$) from the Fe-rich chimneys has high Cu contents (median = 3077 ppm), comparable Co contents (median = 4649 ppm), and low Se (median = 9.19 ppm) contents. The pyrrhotite is generally characterized by significant Co enrichment and a deficit of other trace elements (Fig. 7A).

Isocubanite (CuFe_2S_3): The isocubanite ($n = 33$) in the Zn-rich chimneys has Fe contents of 35.3–43.1 wt.% (median = 40.1 wt.%) and Cu contents of 20.1–27.7 wt.% (median = 23.4 wt.%). In comparison, the isocubanite ($n = 49$) from the Fe-rich chimneys is characterized by higher Fe (median = 42.9 wt.%) and lower Cu (median = 20.1 wt.%) concentrations (Table S1). The LA-ICP-MS analyses show that the isocubanite has higher Co contents than the sphalerite and pyrrhotite (Table 3; Fig. 7A). Among them, the isocubanite ($n = 10$) in the Zn-rich chimneys has high Co and Zn concentrations, with median values of 4361 ppm and 2187 ppm, respectively. The isocubanite ($n = 27$) in the Fe-rich chimneys is characterized by lower Co (median = 3583 ppm), Sn (median = 19.1 ppm), and Ag (median = 9.82 ppm) concentrations. In addition, the median Au content is 0.19 ppm, and the peak in Au content during LA-ICP-MS analysis indicate presence of Au inclusions in the isocubanite (Fig. S3D).

Chalcopyrite (CuFeS_2): The average Cu, Fe, Zn, Co, and Sn contents of these samples ($n = 7$) are 30.6 wt.%, 31.6 wt.%, 2.58 wt.%, 0.06 wt.%, and 0.32 wt.%, respectively (Table 2). Chalcopyrite has the highest Sn contents of all of the sulfide minerals. In comparison, the chalcopyrite in the Fe-rich chimneys ($n = 6$) has relatively high Cu (avg. 32.4 wt.%) and Fe (avg. 32.0 wt.%) contents. The LA-ICP-MS analyses

show that the chalcopyrite ($n = 10$) has a higher Co content (median = 4031 ppm) and lower Sn (median = 45.6 ppm) and Ag (9.75 ppm) contents (Table 3). Native gold inclusions are hosted in individual chalcopyrite particles with Au contents up to 1.19 ppm (Fig. S3B).

Pyrite (FeS_2): The Fe and S contents of the pyrite ($n = 14$) in the Fe-rich chimneys have narrow ranges of 45.67–46.90 wt.% and 52.46–53.43 wt.%, respectively (Table S1). The pyrite ($n = 25$) has lower Co (median = 90.1 ppm), Sn (median = 0.53 ppm), Ag (median = 0.21 ppm) contents (Fig. 7A) and higher Pb (median = 287 ppm), Tl (median = 46.1 ppm) contents compared to the other minerals (Table 3).

4.3. Sulfur isotopes

The in situ SHRIMP-SI sulfur isotope analysis results show that the $\delta^{34}S$ values of the sulfides in the Zn-rich chimneys range from 1.47 to 8.39‰. The $\delta^{34}S$ values of the chalcopyrite and sphalerite range from 4.02 to 8.39‰ (avg. 6.92 ± 1.34 ‰; $n = 16$; Table 4) and from 1.47 to 8.34‰ (avg. 4.77 ± 1.20 ‰; $n = 77$), respectively. Different types of sphalerite have significantly different $\delta^{34}S$ values (Fig. 8A): the $\delta^{34}S$ values of the sphalerite-I, sphalerite-II, and sphalerite-III are 1.47–5.74‰ ($n = 12$), 2.12–8.34‰ ($n = 45$); and 3.29–5.80‰ ($n = 20$), respectively (Table S4).

The Fe-rich chimneys have a wider total sulfur isotope range (0.26 to 17.38‰) than the Zn-rich chimneys. However, the $\delta^{34}S$ value ranges of chalcopyrite (from 2.51 to 5.91‰, avg. 4.68 ± 0.8 ‰; $n = 16$) and sphalerite (from 0.26 to 5.85‰, avg. 3.19 ± 1.55 ‰; $n = 15$) are relatively limited (Fig. 8B; Table 4 and S4). In addition, the pyrrhotite and pyrite in the Fe-rich chimneys have much higher $\delta^{34}S$ values than the chalcopyrite and sphalerite. The $\delta^{34}S$ values of the pyrrhotite and pyrite range from 5.25 to 9.79‰ (avg. 7.69 ± 1.05 ‰; $n = 28$) and from 8.22 to 17.38‰ (avg. 13.79 ± 3.50 ‰; $n = 14$), respectively (Table S4).

5. Discussion

5.1. Ore forming conditions of sulfide chimneys

5.1.1. Zn-rich chimneys

Previous studies have shown that variations in the Fe wt.% and FeS mole % contents of sphalerite can indicate the trends of the temperature, sulfur fugacity (fS_2), and other parameters of an ore-forming fluid (e.g., Hannington et al., 1991; Keith et al., 2014). The sphalerite in the Zn-rich chimneys has the widest range of Fe contents (1.18–17.92 wt.%) in the DVF, but the values between three generations of sphalerite are significantly different (Fig. 5A).

The Sph-I with a porous surface is commonly observed close to the fluid channel, where it replaced the pyrrhotite and with chalcopyrite disease (Fig. 4A). It has been previously suggested that the mechanism for the formation of this texture is most likely the replacement of FeS in the sphalerite by chalcopyrite (Eldridge et al., 1988; Govindarao et al., 2018). The chalcopyrite disease texture also represents the earliest stage of chalcopyrite replacement of the sphalerite (Fig. 4A). The median formation temperature of the Sph-I was calculated to be 304°C (Table S1) using a geothermometer (Czamanske, 1974). Brueckner et al. (2016) suggested that the higher the Fe wt% of the sphalerite is, the more reducing the paragenetic hydrothermal fluid is. The Fe content of the Sph-I is higher than those of the other two types of sphalerite (Fig. 5A), so we speculate that this stage was characterized by a high-temperature reducing hydrothermal fluid. The Sph-I also has the highest median value of FeS mol% among the three types of sphalerite (Fig.5). Thus, it reflects the lowest sulfidation state and fS_2 of the fluid.

Close to the fluid channels, the idiomorphic and homogeneous Sph-II is hosted at the edges of the chalcopyrite/isocubanite (Fig. 4E) or pyrrhotite (Fig. 4D), which formed under the early high-temperature conditions. The median formation temperature

of the Sph-II was calculated to be 248°C (Czamanske, 1974) and it has fluid conditions with an intermediate sulfidation state (Fig. 5B; Einaudi et al., 2003). Notably, its FeS mol% range is the widest, demonstrating the wide variation in the sulfur activity during its mineralization. We also note that there is a wide range in both FeS mol% and $\delta^{34}\text{S}$ values of the Sph-II (Figs. 5B, 8A). Thus, we conclude that the most intense mixing of high-temperature hydrothermal fluid and cold seawater occurred during this long stage of formation of sphalerite (Figs. 4B, E). Previous studies have suggested that a sparse and porous beehive structure leads to a high surface area, significant mixing of seawater within the chimney (Fouquet et al., 1993; Webber et al., 2015), and slow cooling (Webber et al., 2017). Therefore, the beehive structure of the Zn-rich chimneys (Fig. S2C) may be an important factor controlling the formation of the Sph-II.

The colloform sphalerite (Sph-III) is mainly present in the outer layer of the chimneys with a zoned structure (Fig. 4F). The median formation temperature for the Sph-III was calculated (Czamanske, 1974) to be 191°C, which is lower than those of the Sph-I and Sph-II (Fig. 5A; Table S1). The lowest median FeS mol% of Sph-III indicates that the ore-forming fluid was characterized by low-temperature oxidation and high sulfur fugacity (Hannington et al., 1989; Einaudi et al., 2003). The irregular zoning characteristics of the colloform sphalerite (Fig. 6) are associated with rapid temperature, pH and f_{S_2} fluctuations during precipitation (Schaarschmidt et al., 2021). This type of environment leads to widespread dissolution, reprecipitation, and concentric growth zones of the mineral phase in the texture (e.g., Haymon, 1983; Turner & Campbell, 1987; Tivey et al., 1990). We propose that the zonation of the Sph-III further indicates unbalanced fluid conditions, that is, a large amount of seawater in the outer layer of the chimney underwent multiple mixing with the hydrothermal fluid. As the vent fluid underwent conductive cooling or interacted with seawater, the modified fluid could precipitate higher f_{S_2} - f_{O_2} mineral assemblages (goethite + iron hydroxide + opal; Figs. 3, 4L) (e.g., Tivey et al., 1999).

The sulfidation state represents the chemical potential of the sulfur relative to the sulfide mineral assemblage, which increases with decreasing temperature as a function of fS_2 and temperature (Einaudi et al., 2003). There is a gradual decrease in the FeS mol% of the sphalerite in the Zn-rich chimneys, and the fluctuations may be caused by intermittent events such as boiling and local wall rock interactions (Einaudi et al., 2003). It can be seen from Fig. 5B that the sulfidation state of the Zn-rich chimneys had a wide range, i.e., it gradually changed from the low sulfidation of the Sph-I to the intermediate sulfidation state of the Sph-II and Sph-III. This indicates a gradual evolution of the fluid from a higher temperature ($\sim 300^\circ\text{C}$), lower sulfidation state (high R_s), and reducing conditions to a lower temperature ($\sim 150^\circ\text{C}$), higher sulfidation state (low R_s), and more oxidizing conditions. This process may have been driven by a decrease in fluid temperature due to conductive cooling, and by increased fS_2 and fO_2 resulting from mixing with oxidized seawater (Kawasumi & Chiba, 2017).

5.1.2. *Fe-rich chimneys*

The pyrrhotite (Fig. 4G) and chalcopyrite/isocubanite (Fig. 4H) within the channels, as the major sulfide mineral in Fe-rich chimneys, usually indicate a fluid environment with a high-temperature and low fS_2 and fO_2 values (e.g., Lusk & Bray, 2002; Choi et al., 2021). The pyrrhotite with a skeleton texture and the iron hydroxide (Fig. 4L) in the outer layer of the chimney demonstrate that it underwent extensive seawater oxidation in the late stage. The narrow temperature ranges ($207\text{--}321^\circ\text{C}$) and FeS content variations of the sphalerite in the Fe-rich chimneys indicate fluid conditions with a medium-high-temperature and intermediate-low sulfidation state (Fig. 5B; Einaudi et al., 2003).

5.2. *Cobalt and tin enrichment*

The sulfides from DVF are characterized by the enrichment in Co and Sn that are significantly different from other NTO-related SMS deposits (Wang et al., 2021).

5.2.1. Cobalt mineralization

The Co enrichment of SMS deposits (mainly ultramafic-hosted) may be explained by the high Co concentration (91 ppm) of upper mantle rocks (Anderson, 1989; Fouquet et al., 2010) and magmatic stratification leading to a higher Co fertile oxide gabbros in the root zone of the detachment (Patten et al., 2024). The sulfide chimneys in the DVF have higher Co contents (avg. 1759 ppm, $n = 39$) than almost all of the studied SMS deposits along mid-ocean ridges (avg. 38 ppm, $n = 685$) and lower than only a few ultramafic deposits (avg. 2926 ppm, $n = 262$), such as the Rainbow and Ashadze1 deposits (Wang et al., 2021 and references therein). Thus, the extreme Co enrichment of the sulfides from the mafic-hosted DVF suggests a possible Co-rich source in the deep reaction zone of this hydrothermal circulation system.

In the DVF, the predominant Co-containing minerals are pyrrhotite (Co = 875–6578 ppm) and isocubanite (Co = 3181–4618 ppm) (Table 3), and their smooth ablation profiles (Figs. 6B, D) indicate that the Co is uniformly distributed in these Fe-bearing minerals. As a divalent cation, Co^{2+} can replace Fe^{2+} and Zn^{2+} in the lattice (Vaughan & Rosso, 2006; Maslennikov et al., 2009). However, hydrothermal fluid observations (Trefry et al., 1994; Metz & Trefry, 2000) and thermodynamic simulations (Williams-Jones & Vasyukova, 2022) have demonstrated that the fluid temperature, $f\text{O}_2$, and pH are important factors controlling Co enrichment in sulfides. The strongly reducing environment indicated by the abundant pyrrhotite and isocubanite (e.g., Figs. 3C, E) in the DVF is favorable for the leaching and transport of Co from the wall rocks by high-temperature hydrothermal fluids (Hannington et al., 2014; Fouquet et al., 2010). The higher Co concentration (avg. 4162 ppm; $n = 37$) of the isocubanite suggests precipitation from a high-temperature ($\sim 350^\circ\text{C}$) reducing hydrothermal fluid (Herzig et al., 1998; Grant et al., 2018). From the Sph-I to Sph-III, the average Co content of the sphalerite decreases sequentially (Fig. 7B) also indicating that the high-temperature, low sulfidation state, and reducing conditions (Fig. 5B) controlled the degree of Co

enrichment.

Previous studies have proposed that phase separation may systematically affect the metal contents (Koschinsky et al., 2008; Webber et al., 2015; Falkenberg et al., 2021) and associated sulfide precipitation from hydrothermal fluids (Wohlgemuth-Ueberwasser et al., 2015; Keith et al., 2016), which is considered to be one reason responsible for the cobalt enrichment in Daxi, especially in the Fe-rich chimneys (Table 3). Cobalt is often transported as CoCl_4^{2-} in acidic, strongly reducing Cl-rich fluids with temperatures of $>250^\circ\text{C}$ (e.g., Metz & Trefry, 2000; Migdisov et al., 2011). As the affinity of Co for Cl increases with increasing temperature and salt concentration (Brugger et al., 2016), Co tends to be enriched in the Cl-rich high-temperature brine phase fluids during phase separation (Heinrich et al., 1999; Keith et al., 2021). Similar Co-enriching process was reported both by the studies of volcanogenic massive sulfide deposits on land and modern seafloor massive sulfides (Keith et al., 2016; Sahlström et al., 2022).

5.2.2. Tin mineralization

Another remarkable mineralization feature of the DVF is the enrichment of Sn (avg. 967 ppm) (Wang et al., 2021). The Sn content is more than an order of magnitude higher than that of other mafic-hosted SMS deposits and is only comparable to that of a few ultramafic-hosted SMS deposits, such as the Logatchev-2 and Nibelungen sites (Fig. S4B). However, the Sn concentration of the ultramafic rocks (0.52 ppm) is lower than that of mid-ocean ridge basalt (MORB) (1.36 ppm; Anderson, 1989), which may not explain the Sn enrichment of ultramafic deposits. In the DVF, the Sn is mostly hosted in sphalerite and chalcopyrite (Fig. 7A). George et al (2018) suggest that Sn can be remobilized from pre-existing sphalerite and incorporated into the chalcopyrite. The trends of the coordinated variations in Sn with Zn and Fe in the sphalerite (Fig. 6) indicate that the Sn may replace Zn^{2+} and Fe^{2+} . The sulfide chimneys in the Yonaguni

Knoll IV hydrothermal field revealed that Sn replaced Fe as a solid solution via the coupled substitution of $\text{Sn}^{4+}\text{Fe}^{2+}$ for $\text{Fe}^{3+}\text{Fe}^{3+}$ (Gena et al., 2013).

Since Sn is a volatile trace metal with three oxidation states (i.e., Sn^0 , Sn^{2+} , and Sn^{4+}), its geochemical behavior is strongly influenced by the redox conditions (Hayes & McCullough, 2018). It is generally accepted that in reducing hydrothermal fluids, Sn is mainly transported as Sn^{2+} complexes (e.g., Heinrich & Eadington, 1986; Sherman et al., 2000; Migdisov & Williams-Jones, 2005). A strongly reducing environment enhances the mobility and solubility of Sn^{2+} during high-temperature fluid circulation (Schmidt et al., 2011), while Sn^{2+} is converted to Sn^{4+} after mixing with seawater, leading to Sn enrichment in the form of Sn^{4+} (Choi et al., 2021). The Sph-I has a significantly higher Sn content (Fig. 7B, Table S2), suggesting that low sulfidation state and strongly reducing characteristics of the hydrothermal fluid (Fig. 5B) were important factors leading to the Sn enrichment in the DVF.

5.2.3. Evidence from sulfur isotopes

Sulfur isotopes can be used to distinguish fluid sources and reflect mineralization processes (Shanks et al., 2001). In the sediment-starved environment of mid-ocean ridges, the sulfur in hydrothermal sulfides is mainly derived from magmatic sulfides ($\delta^{34}\text{S} = -0.9$ to -0.6% , Labidi et al., 2012, 2014) and seawater sulfates ($\delta^{34}\text{S} = 21.24\%$, Tostevin et al., 2014). There is also significant contribution of magmatic degassing and microbial activity in some areas, with H_2S from these two sources typically exhibiting negative S isotopic composition (Herzig et al. 1998; Shanks et al., 2001). However, no clear signature in Daxi has been observed to support the presence of sulfur from these two sources. In high-temperature hydrothermal systems, the processes of seawater mixing and the thermochemical sulfate reduction (TSR) are important process leading to the increase in the $\delta^{34}\text{S}$ value, which occurs through reduction of substances such as Fe^{2+} , H_2 and CH_4 (Shanks et al., 1981; Meshoulam et al., 2016).

The overall $\delta^{34}\text{S}$ value of DVF exhibits a relatively heavy S isotopic composition ($\delta^{34}\text{S}_{\text{avg}} = 6.1\text{‰}$), which is comparable to tectonically-controlled hydrothermal fields such as TAG ($\delta^{34}\text{S}_{\text{avg}} = 6.8\text{‰}$) and Logatchev ($\delta^{34}\text{S}_{\text{avg}} = 5.7\text{‰}$), but heavier than magmatic-controlled systems like Lucky Strike ($\delta^{34}\text{S}_{\text{avg}} = 2.4\text{‰}$) and Snake Pit ($\delta^{34}\text{S}_{\text{avg}} = 2.1\text{‰}$) (Petersen et al., 2000; Zeng et al., 2017 and references therein). Cherkashev et al. (2003) proposed that these differences are determined by two genetically different types of hydrothermal systems, shallow- and deep-seated systems. The enrichment of heavy sulfur isotopes in tectonically-controlled systems, on the one hand, is typically associated with deep-seated sulfate reduction along faults that penetrate deeply into the crust and with the prolonged residence time of fluids within the hydrothermal system (Petersen et al., 2000; Zeng et al., 2017; Ding et al., 2020). DVF is located within the NTO tectonic setting, where the presence of deep-seated faults is plausible. On the other hand, serpentinization of ultramafic rocks within some tectonically-controlled systems could provide high amounts of hydrogen (H_2) and methane, which contributes to thermochemical sulfate reduction (Bach et al., 2006; Alt et al., 2007; Peters et al., 2010; Alt & Shnaks, 2011). Although no ultramafic rocks have been observed in the Daxi field, high concentrations of dissolved hydrogen (H_2 , 7.07 mM) and methane (CH_4 , 0.884 mM), which suggests fluid and ultramafic rock interactions, have been detected in the vent fluid (Wu et al., 2022). Thus, the abundance of reducing agents, together with potential for deep-seated seawater sulfate reduction, contributes to the relatively heavy S isotopic composition of DVF.

Significant differences in the $\delta^{34}\text{S}$ values are observed among various minerals in the chimneys of Daxi, indicating diverse geological controls. The S isotopic composition of different sphalerite phases in the Zn-rich chimneys varies with the evolution of the hydrothermal fluid. The Sph-II ($\delta^{34}\text{S}_{\text{avg}} = 5.08\text{‰}$) and Sph-III ($\delta^{34}\text{S}_{\text{avg}} = 4.79\text{‰}$), formed at intermediate to low temperatures, show heavier $\delta^{34}\text{S}$ values than high temperature Sph-I ($\delta^{34}\text{S}_{\text{avg}} = 3.59\text{‰}$) (Figs. 7A). This suggests an increased contribution of seawater-derived sulfur in the system at intermediate to low

temperatures. In the Fe-rich chimneys, the pyrrhotite has significantly heavier $\delta^{34}\text{S}$ values ($\delta^{34}\text{S}_{\text{avg}} = 7.69\text{‰}$) than sphalerite ($\delta^{34}\text{S}_{\text{avg}} = 3.19\text{‰}$) and chalcopyrite ($\delta^{34}\text{S}_{\text{avg}} = 4.68\text{‰}$). The mineralogy reveals that the early-formed pyrrhotite is overgrown by late-formed chalcopyrite and sphalerite (Figs. 4G, I). Since the sulfur isotope fractionation factor between minerals and H_2S is typically less than 1‰ at the temperatures of black smokers (Ohmoto & Goldhaber, 1997; Bluth & Ohmoto, 1988; Shanks, 2001), the observed differences in sulfur isotope composition between the minerals cannot be solely explained by fractionation effects. We propose that the variations are related to changes in fluid composition during mineral precipitation. The oceanic crust is initially saturated with seawater and sulfate precipitated before hydrothermal activity. Hydrothermal fluids ascending along conduits will react with the precipitated sulfates and any seawater still percolating through the crust (Petersen et al., 2000). This process leads to partial or complete reduction of the sulfate, resulting in the inheritance of a heavier $\delta^{34}\text{S}$ ratio in the early-formed pyrrhotite. As the hydrothermal activity progressed and the water-rock ratio decreased, H_2S formed through seawater sulfate reduction mixed with the H_2S produced by dissolution of magmatic sulfides (Petersen et al., 2000; Shanks, 2001). This resulted in late-formed chalcopyrite and sphalerite with overall lighter $\delta^{34}\text{S}$ values compared to pyrrhotite.

The pyrite formed in sulfide veins within the fractures of breccias (Fig. 4K) exhibits the highest reported $\delta^{34}\text{S}$ value ($\delta^{34}\text{S}_{\text{max}} = 17.38\text{‰}$) reported to date for hydrothermal sulfides along mid-ocean ridges. Similar heavy sulfur isotope enrichment has been observed at Rainbow hydrothermal system ($\delta^{34}\text{S}_{\text{max}}=12.5\text{‰}$; Bogdanov et al., 2002) and Tianzuo hydrothermal system ($\delta^{34}\text{S}_{\text{max}}=14.1\text{‰}$; Ding et al., 2020). According to the geochemical reaction modeling, thermochemical reduction of seawater sulfate in the open system can at most produce pyrite with $\delta^{34}\text{S}$ values up to approximately 13.5‰ (Alt et al., 2003). In other words, such a process is unlikely to account for the anomalously heavy sulfur isotope values observed in the pyrite. Thus, this anomalously high ^{34}S enrichment is inferred to be associated with Rayleigh

fractionation during TSR processes in closed systems (Hoefs, 2018; Dubinina & Bortnikov, 2019; Dubinina et al., 2020). Although hydrothermal systems are often dynamic with fluids entering and exiting through fractures and permeability pathways, a hydrothermal system or a localized structure (like mound and chimney) can be considered closed if the reaction rate of seawater sulfate with reducing agents significantly exceeds the seawater sulfate supply rate to the system (Peter & Shanks, 1992; Dubinina & Bortnikov, 2019; Dubinina et al., 2020). For the entire hydrothermal system, closed conditions are possible as previous studies indicate that the duration of fluid filtration in the hydrothermal cell (a few years to decades; Kadko et al., 1985) is much longer than the typical time of seawater sulfate reduction and attainment of the isotope equilibrium between sulfate and sulfide in fluid (a few hours to tens of days; Shanks et al., 1981; Ohmoto & Lasaga, 1982; Dubinina & Bortnikov, 2019). Furthermore, for the pyrite within the breccia fractures at the base of the chimney (Fig. 4K), which is enclosed by dense rock breccias and chimney wall and unable to directly contact with abundant seawater, the rate of sulfate reduction is likely to exceed the rate of seawater sulfate supply. In this context, we propose that the pyrites, with anomalous enrichment of heavy sulfur isotope, is influenced by the Rayleigh fractionation effects occurs in the hydrothermal cell or in the special localized structure.

5.3. Genetic implications of mineralization at non-transform offsets

It is commonly assumed that the magmatic/volcanic activity is sparse and the tectonic activity is intense at the end of the second-order segments (Tolstoy et al., 1993; Devey et al., 2010). The DVF is located on a neo-volcanic ridge in a NTO at the ends of two second-order ridge segments along the Carlsberg Ridge (Fig. 1B). It differs from the majority of NTO-associated systems, which are mainly hosted by mantle rocks peridotite (e.g., Rainbow), whereas the host rocks of the DVF are basaltic, including abundant sheet flows and pillow basalts (Wang et al., 2021). However, the enrichment of Co and Sn in the chimneys (Fig. 7) and CH₄ and H₂ in

the fluids (Wu et al., 2022) in the DVF suggests that the circulation of fluids was influenced by the ultramafic substrate. Similar fluid circulation processes have been reported in other seafloor hydrothermal fields, for example, Beebe and Aurora vent fields, which all occur in topographic highs in the thin oceanic crust (Webber et al., 2017; German et al., 2022). The Beebe Vent Field is located on a very thin oceanic crust (2–3 km), where fluid circulation reaches a depth of 1.8 km and interacts with ultramafic lithologies in the basement (Webber et al., 2015). Similarly, the hydrothermal convection cell in the Aurora hydrothermal field could access the ultramafic rocks (Edmonds et al., 2003) under the anomalously thin oceanic crust (~1.9–3.3 km thick; Jokat et al., 2003) and the hydrothermal fluid have a high CH₄:Mn ratio (~1:1; German et al., 2022).

These similarities suggest that a comparable thin oceanic crust also underlies the DVF (see Fig. S1), through which seawater infiltrated via normal faults into the graben's base, where it was heated, sustaining the extended hydrothermal circulation in the DVF. The enrichment of heavy sulfur isotopes in tectonically controlled systems is commonly linked to deep-seated sulfate reduction along faults penetrating deep into the crust and the prolonged residence time of fluids in a hydrothermal system (Petersen et al., 2000; Zeng et al., 2017). The overall $\delta^{34}\text{S}$ value of DVF exhibits a relatively heavy S isotopic composition ($\delta^{34}\text{S}_{\text{ave}} = 6.1\text{‰}$), which is comparable to systems influenced by tectonic processes such as TAG ($\delta^{34}\text{S}_{\text{ave}} = 6.8\text{‰}$; Petersen et al., 2000) and Logatchev ($\delta^{34}\text{S}_{\text{ave}} = 5.7\text{‰}$), but heavier than other magmatic controlled systems, like Lucky Strike ($\delta^{34}\text{S}_{\text{ave}} = 2.4\text{‰}$) and Snake Pit ($\delta^{34}\text{S}_{\text{ave}} = 2.1\text{‰}$) (Zeng et al., 2017 and references therein). The widespread fractures and faults in NTO enhance the permeability for seawater ingress (Devey et al., 2010). Therefore, we propose that this unique geological setting may lead to relatively complex mineralization process (Patten et al., 2021; Albers et al., 2025).

The hydrothermal fluids in the high-temperature reaction zone reacted with the

olivine in the ultramafic lithologies at depth, effectively leaching cobalt and leading to the enrichment of H_2 and CH_4 during serpentinization. This phenomenon is attributed to the elevated concentrations of Co in ultramafic rocks and/or extreme reducing conditions (Fouquet et al., 2010; Patten et al., 2021; Sahlström et al., 2022). Upward-flowing reducing fluids efficiently leached tin from the basaltic oceanic crust, which contains higher tin concentrations (1.36 ppm) than the mantle (0.79 ppm) (Anderson, 1989), potentially contributing to the enrichment of tin in the chimney. Choi et al. (2023) suggested that the low redox potential of hydrothermal fluids in ultramafic-hosted systems could play a crucial role in facilitating the transport of $SnCl_2$ (Sn^{2+}) during hydrothermal circulation. Similar to the Fåvne Vent Field at the ultraslow-spreading Mohs Ridge (Gini et al., 2024), the potential Cu-rich sulfides in the seafloor of the DVF may have precipitated due to the entrainment of seawater (Wang et al., 2021; Zhao et al., 2025), which is consistent with the low in Cu and high in Zn in the hydrothermal fluid (Wu et al., 2022) and sulfide chimney samples (Fig. S4A). The extreme depletion in Ba (avg. 2 ppm; Wang et al., 2021) and lack of barite within the chimneys further suggest seafloor fluid mixing due to the very low solubility of barite in the presence of dissolved sulfate from infiltrated seawater (Seyfried et al., 2003; Jamieson et al., 2016). From a mineralization perspective, the NTO setting appears to favor the formation of economically interesting deposits.

6. Conclusions

The Daxi Vent Field is located on a rifted volcanic ridge within an NTO on the slow-spreading Carlsberg Ridge. Compare to other mafic-hosted SMS deposits, the DVF is characterized by stronger Co (avg. 1759 ppm) and Sn (avg. 967 ppm) enrichment, with heavy $\delta^{34}S$ in pyrite (up to 17.38 ‰). We conclude that the unusual Co and Sn enrichment as well as unique S isotopic composition in the DVF are linked to the thin ocean crust at NTOs, where the hydrothermal fluid can be circulated to a sufficient depth and affected by ultramafic rocks. The strongly reducing fluids could

efficiently leach Co from the ultramafic lithologies at depth and further enrich Sn as it flows upward through the basalts. Furthermore, the highly saline fluids produced by the phase separation process led to strong enrichment of Co in the pyrrhotite and isocubanite whereas fluid-seawater mixing are essential processes contributing to the Sn concentrations of the sphalerite. The abundance of reducing agents, together with potential for deep-seated seawater sulfate reduction, contributes to the relatively heavy S isotopic composition of DVF. The precipitation of pyrite in a relatively closed environment intensified by the Rayleigh fractionation, leading to significant enrichment of ^{34}S . This study expands our understanding of the diversity of hydrothermal mineralization at slow-spreading ridges and assesses the genetic implications of mineralization associated with neo-volcanic ridges at NTOs.

Acknowledgments

This study is funded by National Key Research and Development Program of China (2021YFF0501304, 2018YFC0309903), the National Natural Science Foundation of China (41976076), and China Ocean Mineral Resources R&D Association project (DY135-S2-1-05). We are grateful to the submersible *Jiaolong* operations team from China National Deep Sea Center for their technical support of field investigation. We appreciate Dr. Jihao Zhu for their assistance with the EPMA analyses at SIOMNR and Mrs. Sabine Lange for her assistance with the LA-ICP-MS analyses at GEOMAR. We also thank Dr. Bin Fu from the SHRIMP Laboratory of the Australian National University for his help with the isotope analysis. Dr. John Jamieson is thanked for his valuable discussions on this manuscript. We thank Dr. Simon Hector and another anonymous reviewer for their constructive reviews. The authors also thank Editor-in-Chief Prof. Marco Fiorentini for the efficient editorial handling of the manuscript.

Data availability

All the geochemical and isotopic data is included in this paper and in the Supporting Information and are archived at Zenodo (Wang et al., 2024). The topographic figures were made with Generic Mapping Tool (GMT) version 6.0 (Wessel

et al., 2019). Bathymetric data from GEBCO Compilation Group (2019) was used for mapping the location of seafloor hydrothermal fields along the slow- and ultraslow-spreading ridges. The mantle Bouguer anomaly was calculated following standard procedures used at mid-ocean ridges (Kuo & Forsyth, 1988) using a 1 min satellite-derived free air gravity anomalies (Sandwell et al., 2014) and the compiled multibeam bathymetry grid (Wang et al., 2021).

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Journal Pre-proof

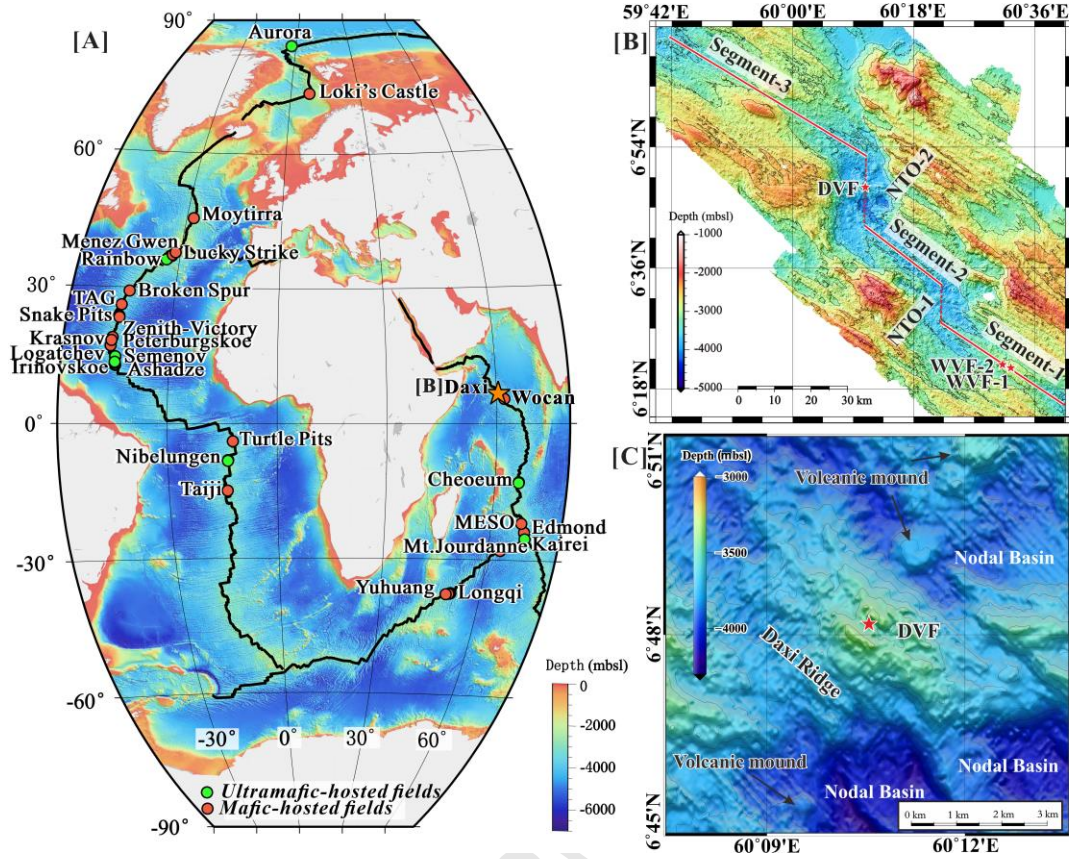


Fig. 1. (A) Typical seafloor hydrothermal fields with massive sulfide deposits along slow- and ultraslow-spreading ridges. (B) Bathymetric map of the Daxi-Wocan segments (6–7 °N) of the slow-spreading Carlsberg Ridge showing the location of the Daxi Vent Field (modified from Wang et al., 2021). (C) Close up of the Daxi Vent Field showing its location on top of the rifted Daxi volcanic ridge at NTO-2 near 6°48'N (modified from Wang et al., 2021). Note: CR - Carlsberg Ridge; CIR - Central Indian Ridge; SWIR - Southwest Indian Ridge; MAR - Mid-Atlantic Ridge; AMOR - Arctic Mid-Ocean Ridge.

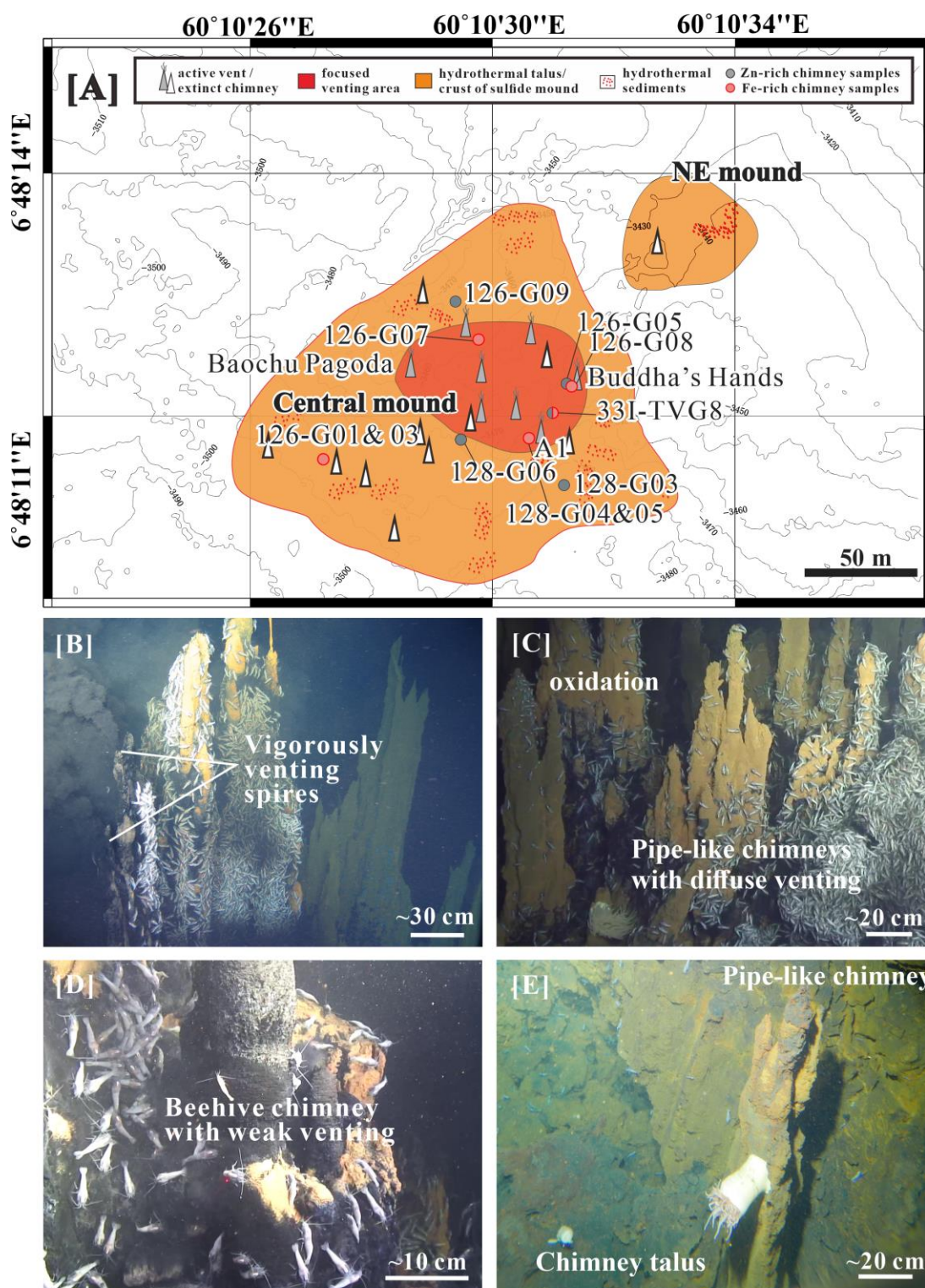


Fig. 2. Distribution of sampling sites and examples of chimney and mineralization types encountered during manned submersible dive operations. (A) An interpretive map of the Daxi Vent Field based on manned submersible videos and sampling (modified from

Wang et al., 2021), with sample locations indicated. (B) Vigorously venting spires on the top of a ~24 m tall chimney, field of view ~5 m. (C) Pike-like chimneys with diffuse venting and a large amount of shrimp aggregation, field of view ~5 m. (D) Beehive-like chimney with weak venting, field of view ~1 m. (E) Pipe-like inactive chimney with minor oxidation on the central mound, field of view ~2 m.

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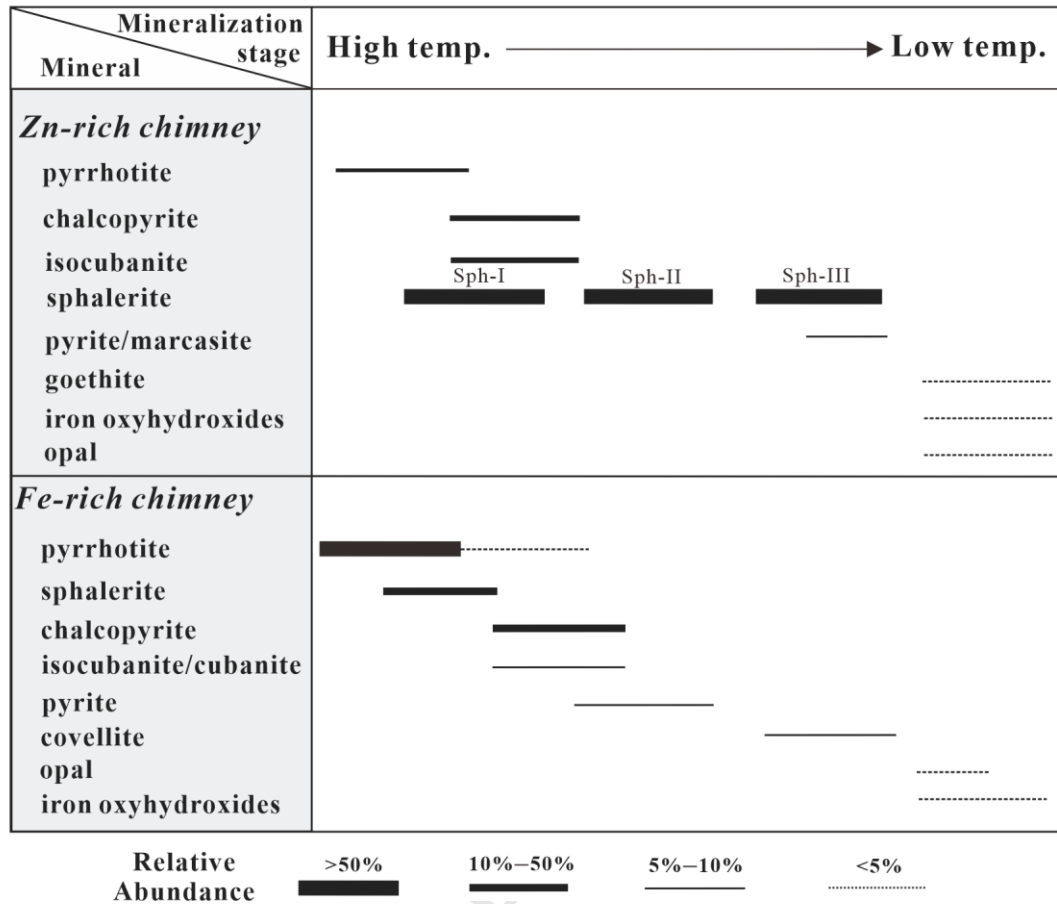


Fig. 3. Interpreted paragenetic sequence of ore and gangue minerals in the two types of sulfide chimneys identified in the Daxi Vent Field.

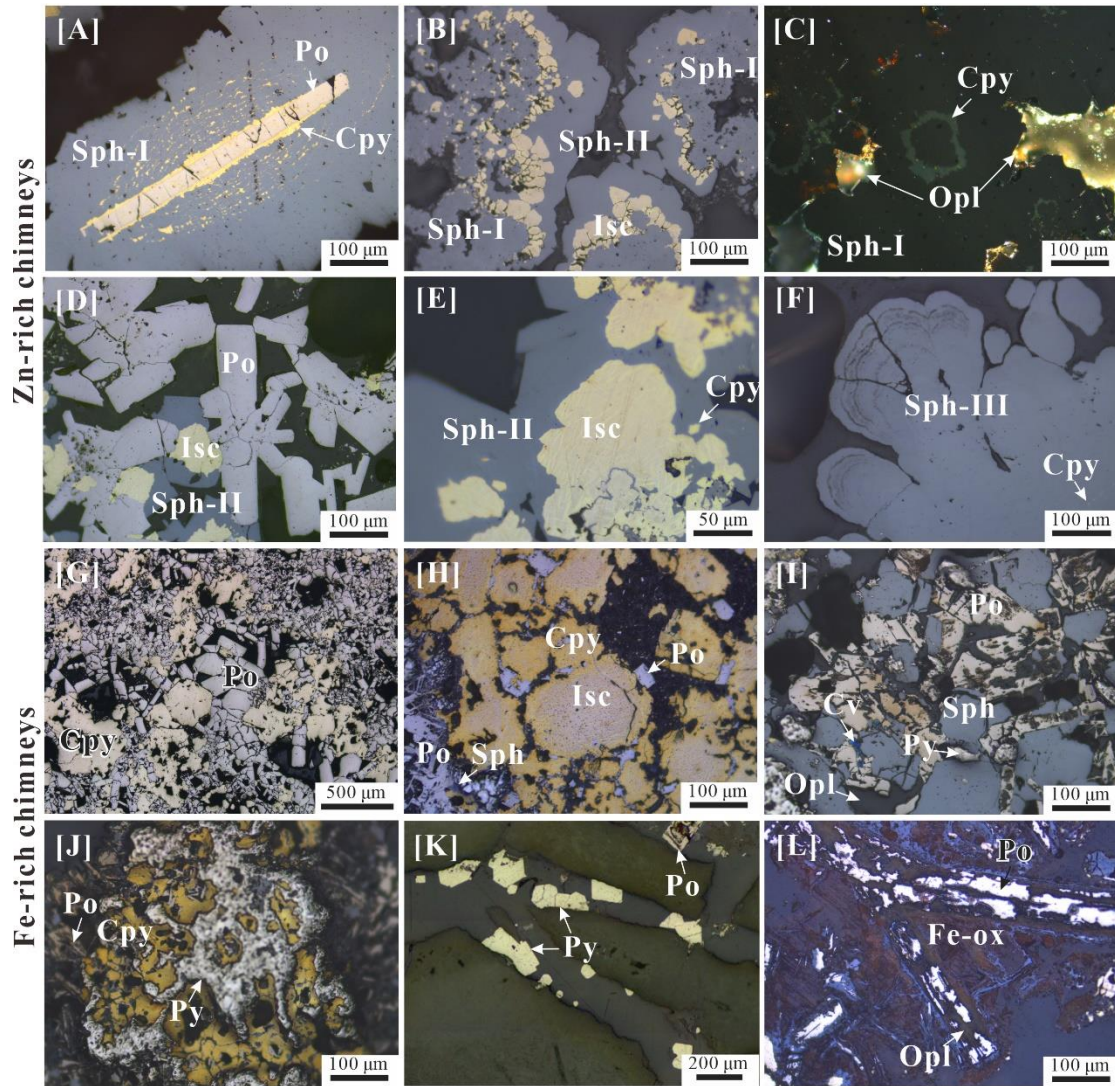


Fig. 4. Microscopic hydrothermal features of selected chimney samples from the Daxi Vent Field, shown by (A–L) photomicrographs in reflected- or transmitted-light. (A) Pyrrhotite crystal partially replaced by chalcopyrite and sphalerite (TVG18C); (B) Isocubanite mineralized between Sph-II and Sph-I (sample JL126-G05); (C) Opal (Opl) cemented sphalerite (TVG18C); (D) Interstitial isocubanite associated with idiomorphic lath shaped pyrrhotite (Po) (JL128-G03); (E) Isocubanite (Isc) with chalcopyrite exsolution lamellae, intimately intergrown with sphalerite in the interior conduit of a Zn-rich chimney (JL128-G03); (F) Colloform sphalerite (Sph-III) with chalcopyrite (Cpy) disease in the exterior wall of a Zn-rich chimney (JL128-G03); (G) Chalcopyrite filling interstices between laths of pyrrhotite in the fluid channel in a Fe-

rich chimney (JL126-G03); (H) Association of isocubanite, sphalerite, chalcopyrite, and pyrrhotite (JL128-G05); (I) Fe-rich chimney with pyrrhotite, sphalerite, pyrite, and covellite (Cv) (JL126-G07); (J) Chalcopyrite mantled by late pyrite (Py) (JL126-G07); (K) Subhedral pyrite form sulfide veins in the fractures of breccias (JL128-G04); (L) Pyrrhotite that has been strongly altered and partially replaced by iron oxyhydroxides (Fe-ox) in the exterior wall of an Fe-rich chimney (JL128-G04).

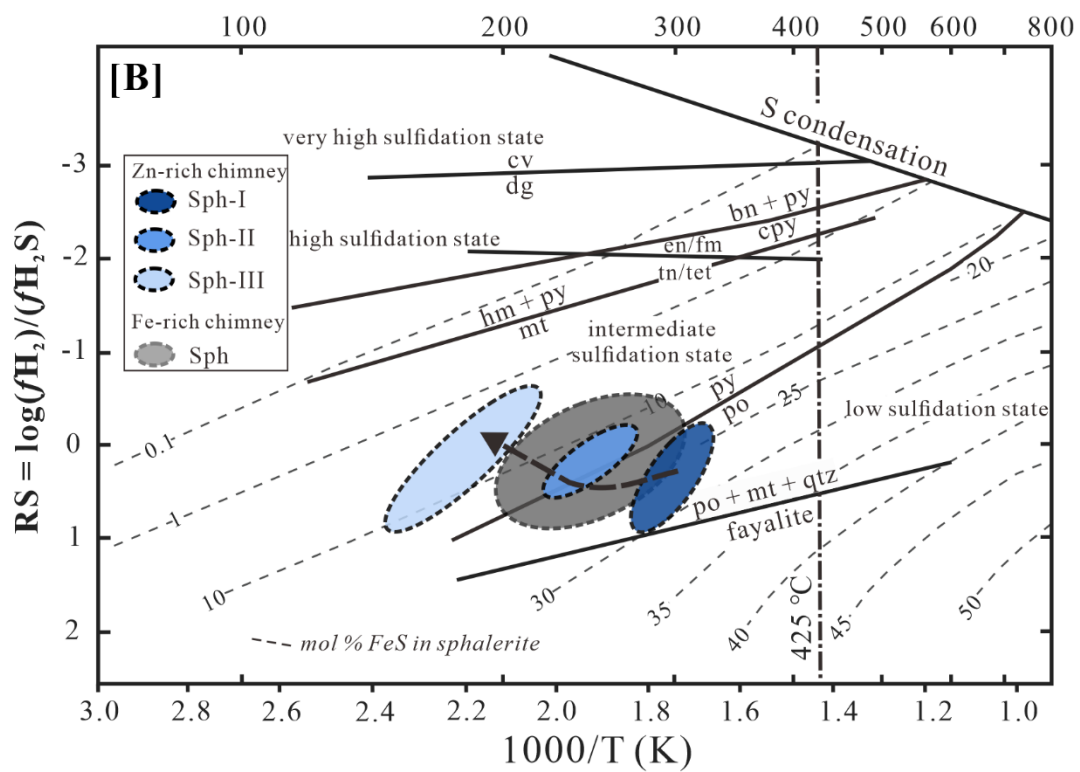
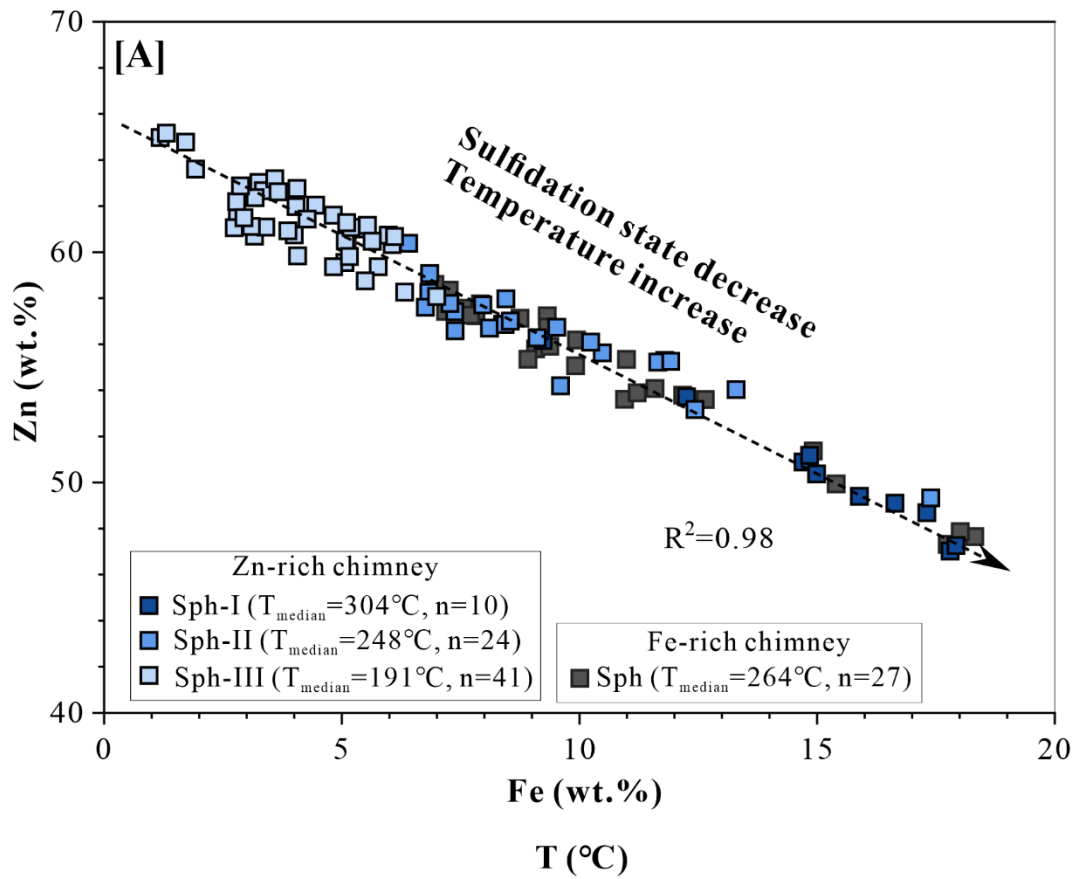


Fig. 5. (A) Binary plots Fe vs. Zn (EPMA major element data) ratios for sphalerite ($n = 102$) from the two types of chimneys in the Daxi Vent Field. The plot reveals the decreasing trend of the sulfidation state (Keith et al., 2014). (B) Plot of temperature vs sulfidation potential (RS) diagram showing the varying Fe contents of sphalerite coexisting with pyrite – pyrrhotite plot of sulfide chimneys in the Daxi Vent Field (modified from Einaudi et al., 2003). Mineral abbreviations: asp - arsenopyrite, bn - bornite, cpy - chalcopyrite, cv - covellite, dg - digenite, en - enargite, fm - famatinite, hm - hematite, lo - lollingite, mt - magnetite, po - pyrrhotite, py - pyrite, qtz - quartz.

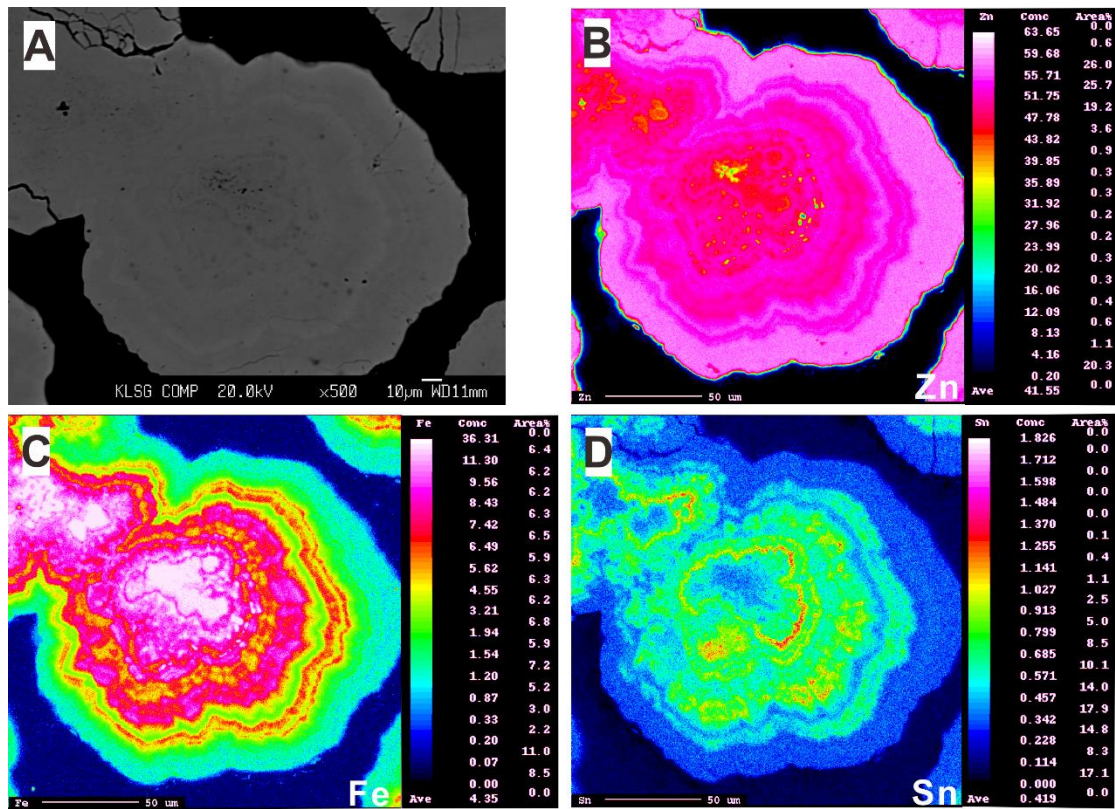


Fig. 6. EPMA-WDS element maps of typical colloform Sn-bearing sphalerite from a Zn-rich chimney (sample JL128-G03).

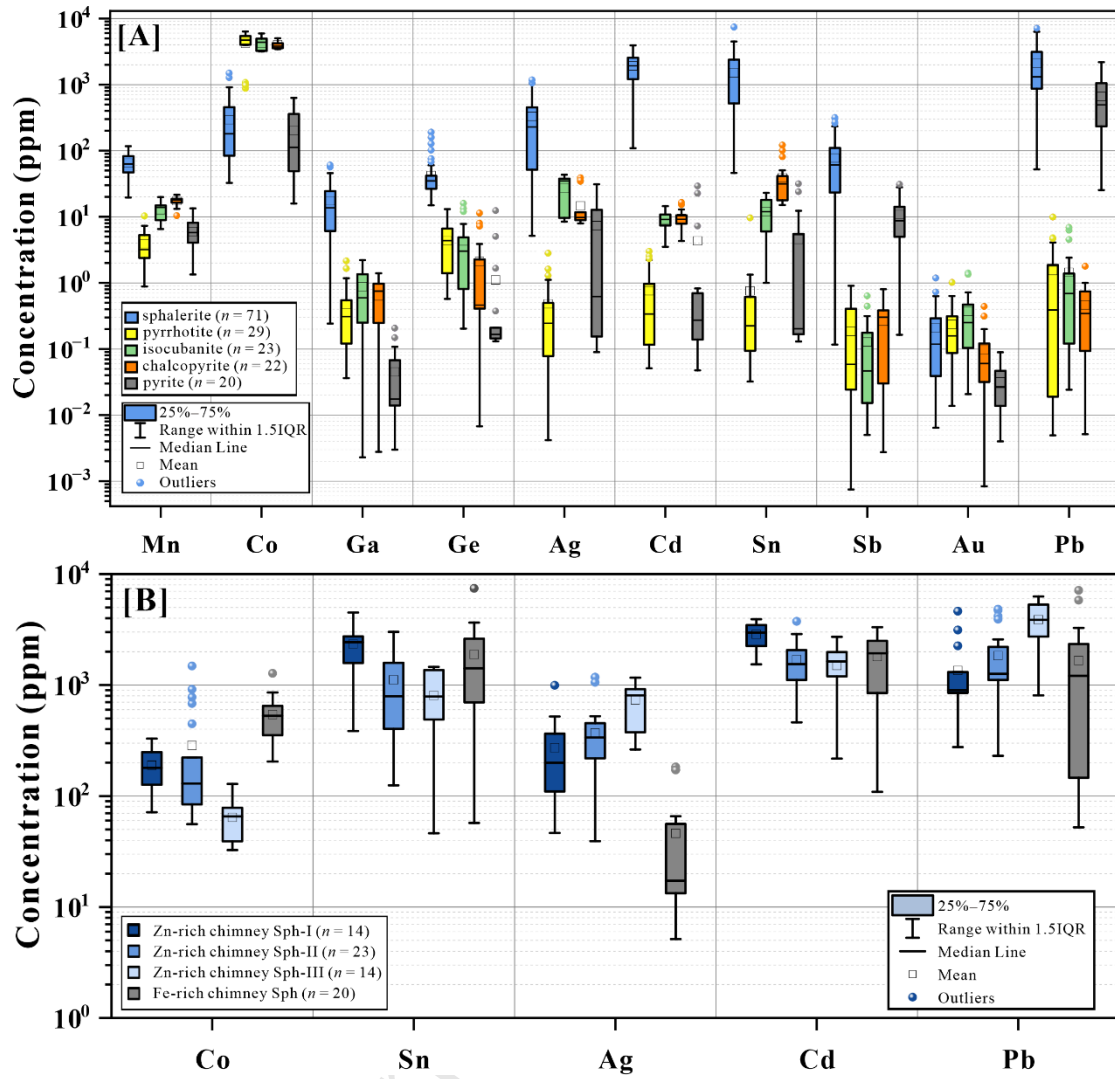


Fig. 7. (A) Box plot of selected trace element compositions of the sulfide minerals from the Daxi Vent Field analyzed via LA-ICP-MS; (B) Selected element distributions of sphalerite from the Zn-rich and Fe-rich chimneys. The full datasets are presented in Table 3.

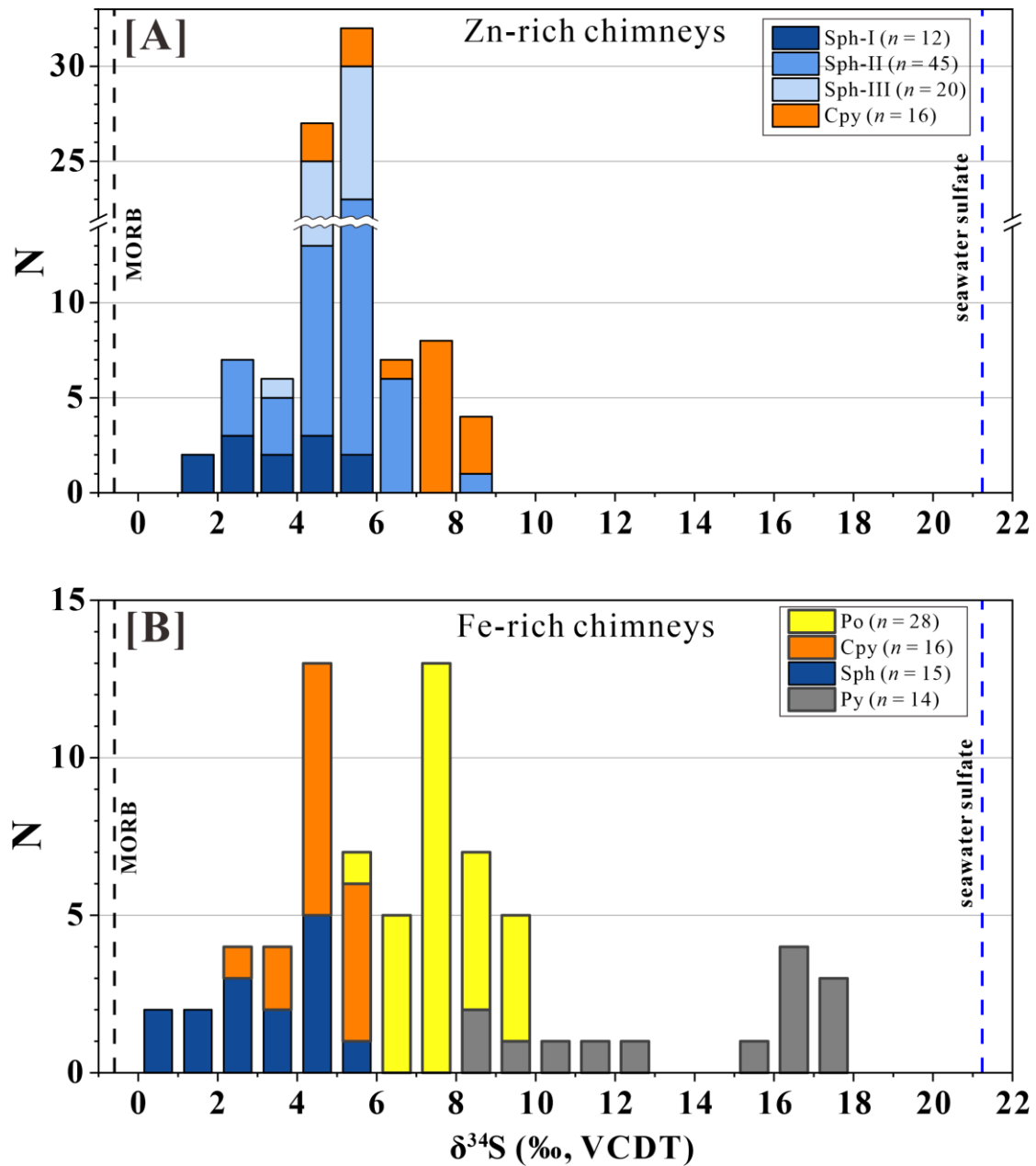


Fig. 8. Frequency distribution of $\delta^{34}\text{S}$ values of sulfide minerals from (A) Zn-rich chimneys and (B) Fe-rich chimneys in the Daxi Vent Field. The sulfur isotope composition of MORB ($\delta^{34}\text{S} = -0.9$ to -0.6‰ ; Labidi et al., 2012, 2014) and seawater sulfate ($\delta^{34}\text{S} = 21.24\text{‰}$; Tostevin et al., 2014) are denoted by the black and blue dashed lines, respectively. The complete sulfur isotope data are presented in Table S4. Abbreviations: Cpy - chalcopyrite, Sph - sphalerite, Py - pyrite, Po - pyrrhotite.

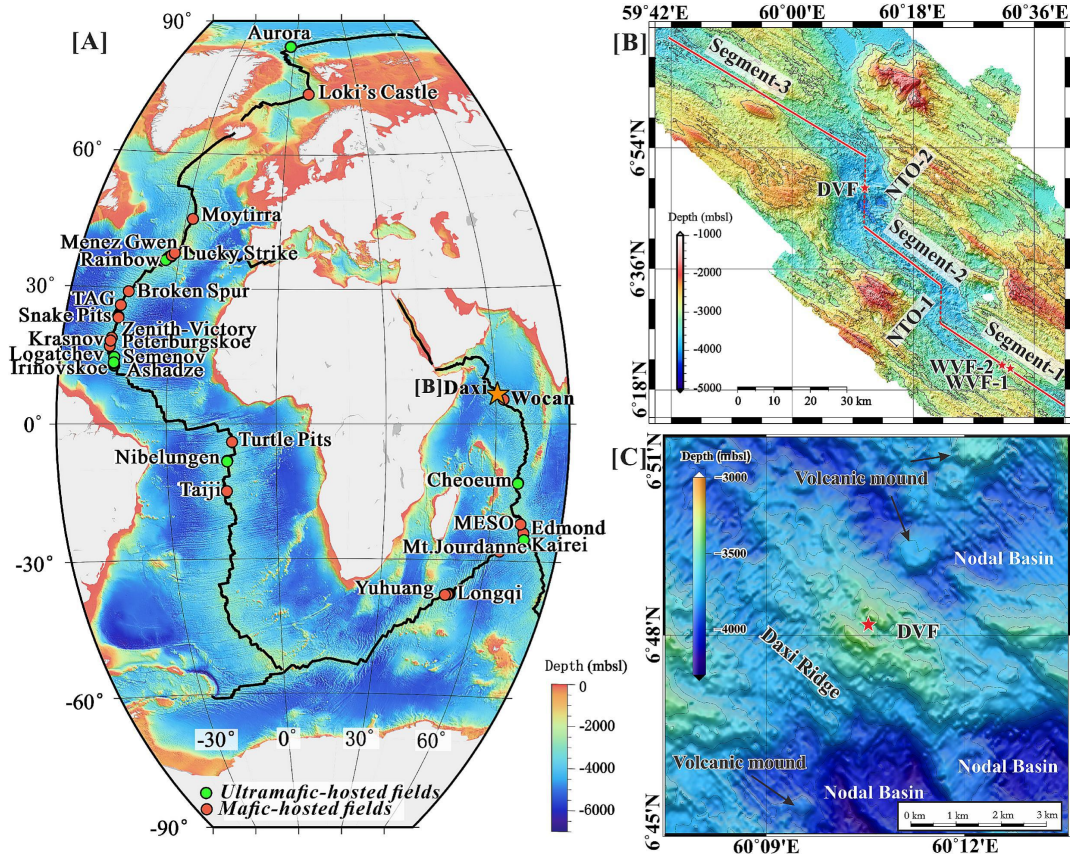


Figure 1

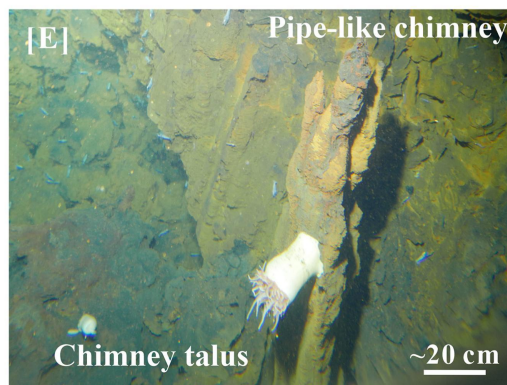
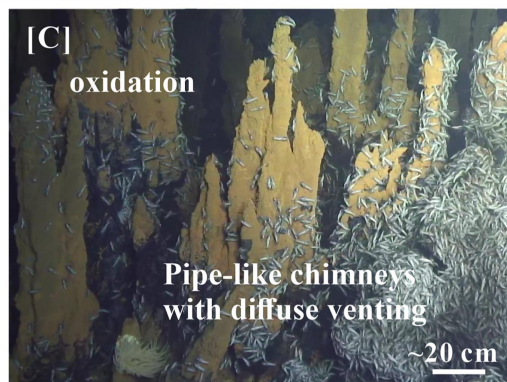
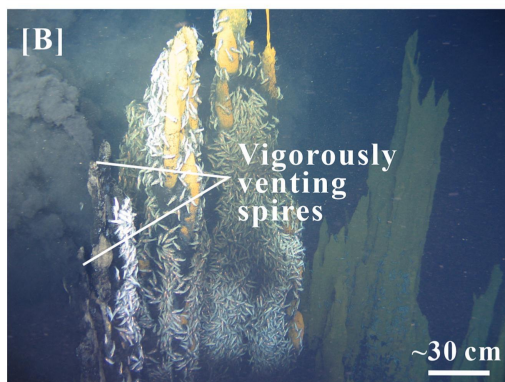
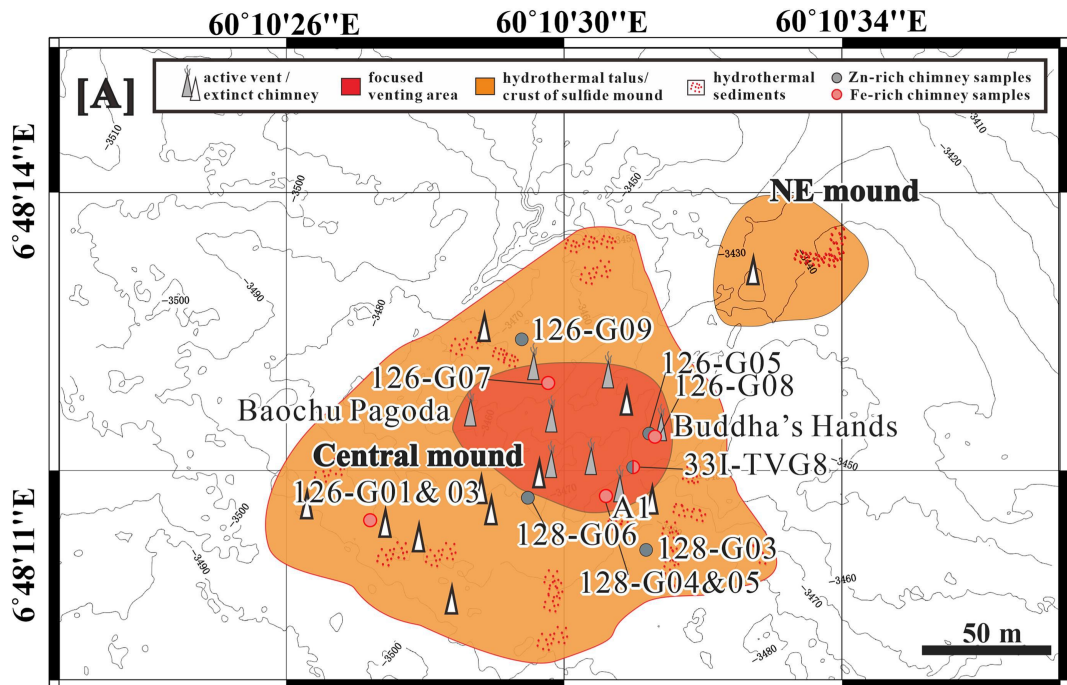


Figure 2

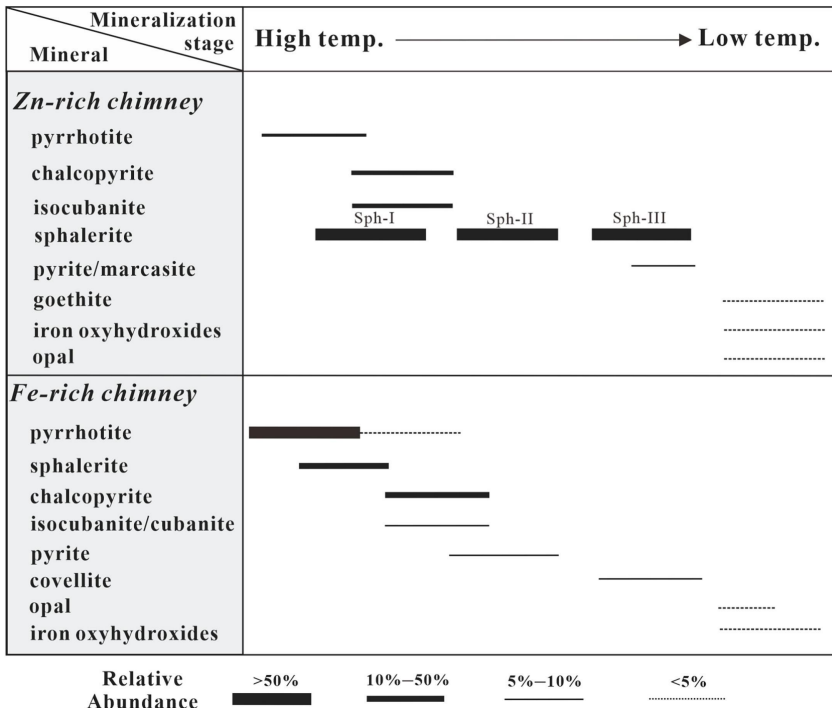
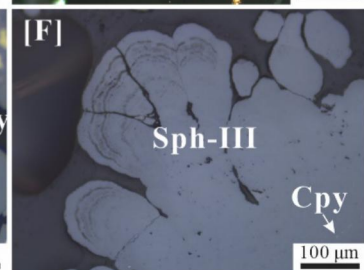
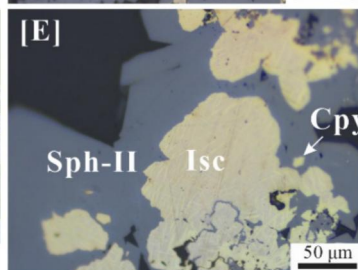
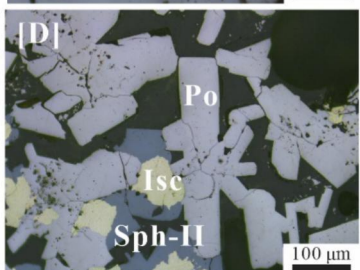
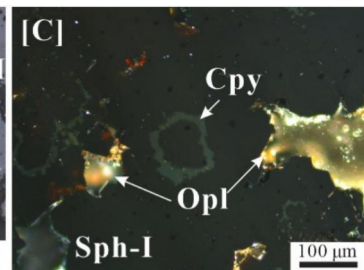
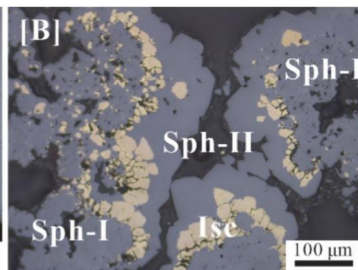
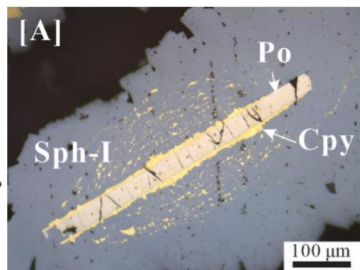


Figure 3

Zn-rich chimneys



Fe-rich chimneys

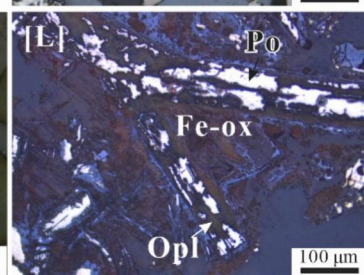
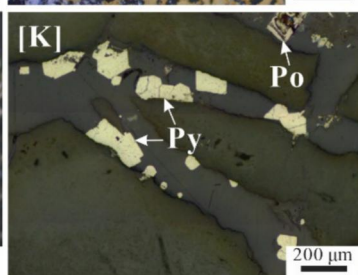
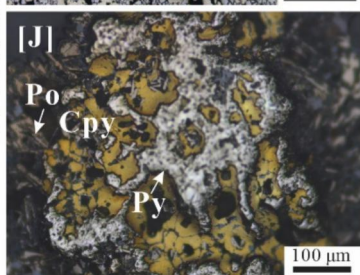
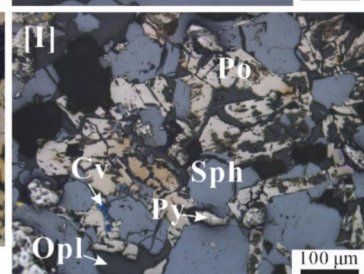
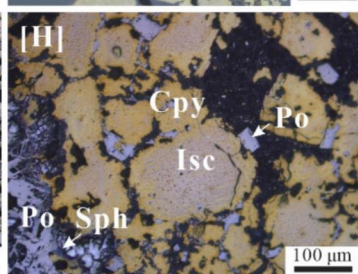
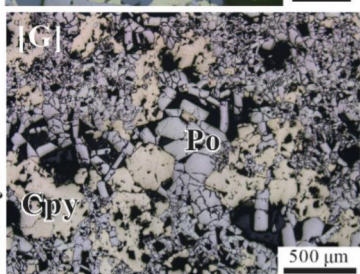


Figure 4

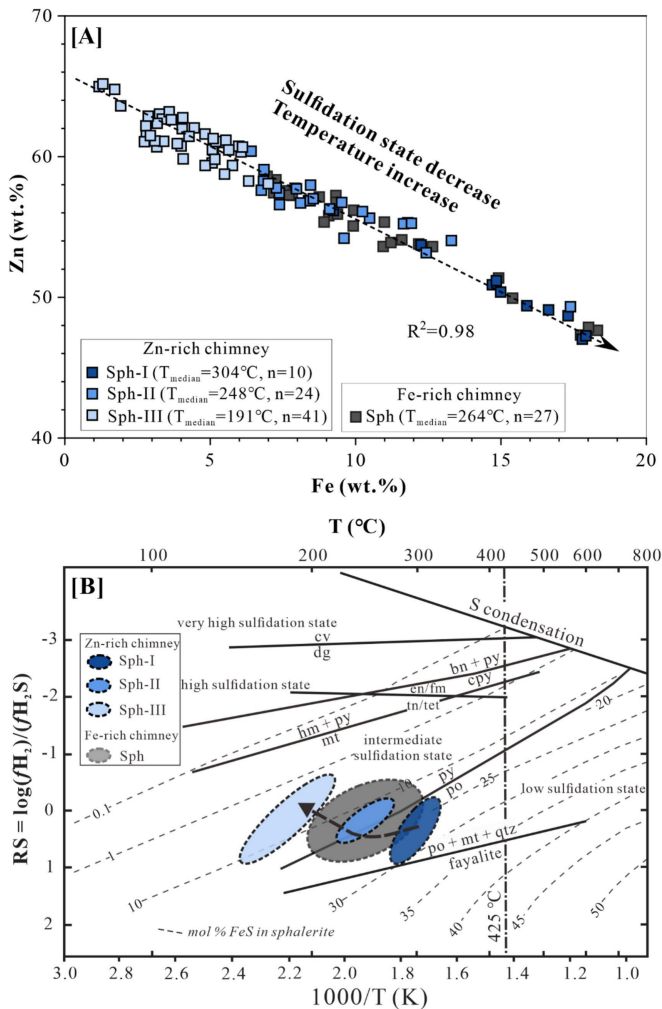


Figure 5

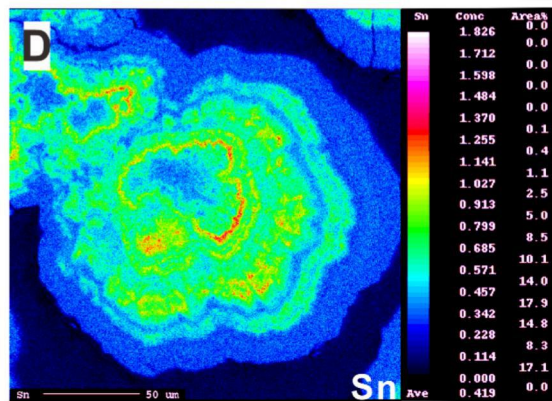
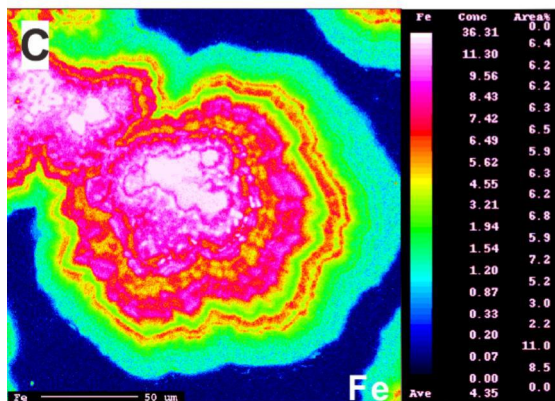
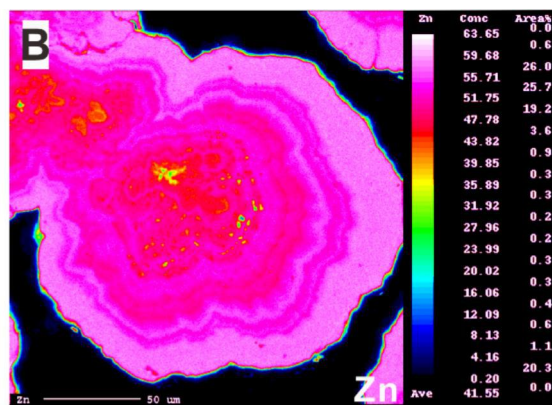
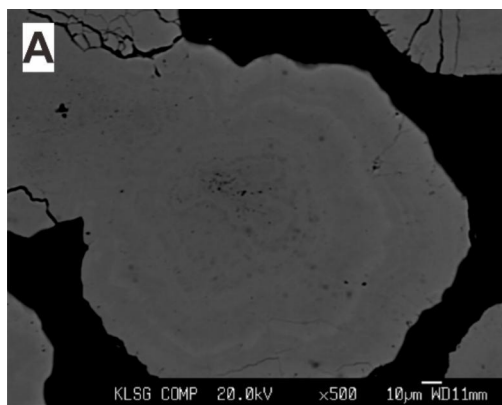


Figure 6

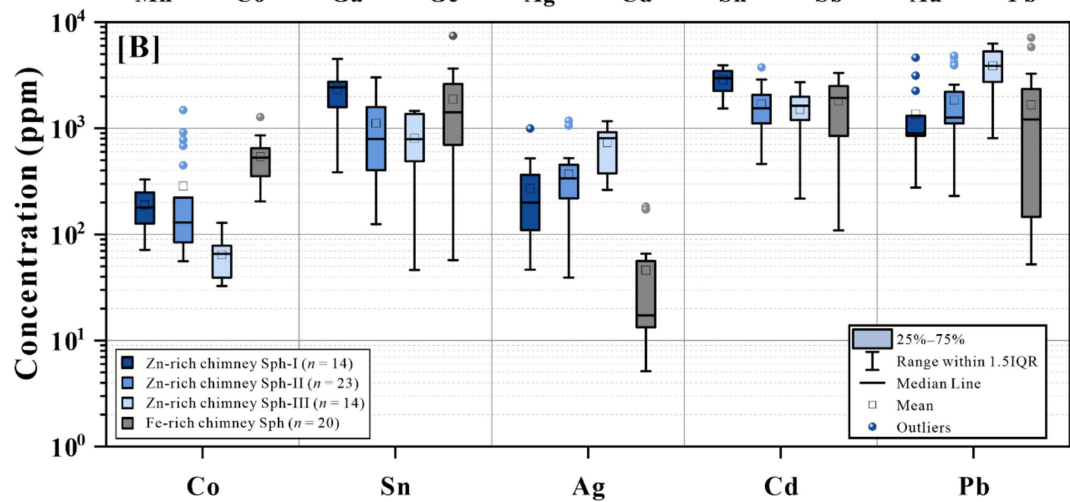
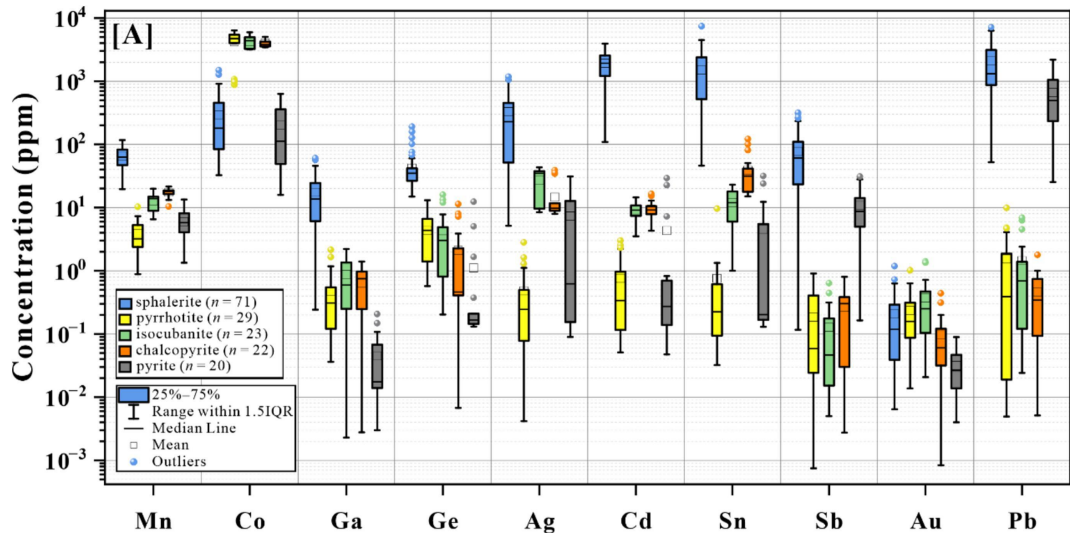


Figure 7

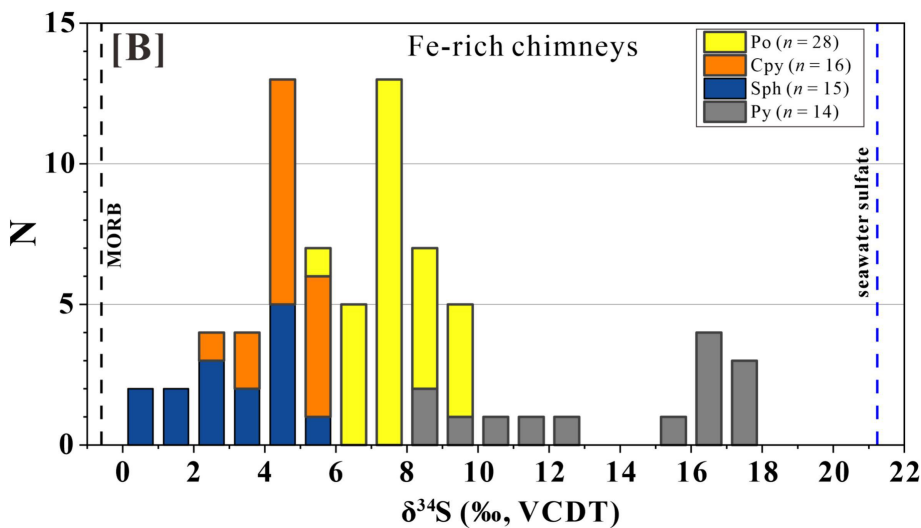
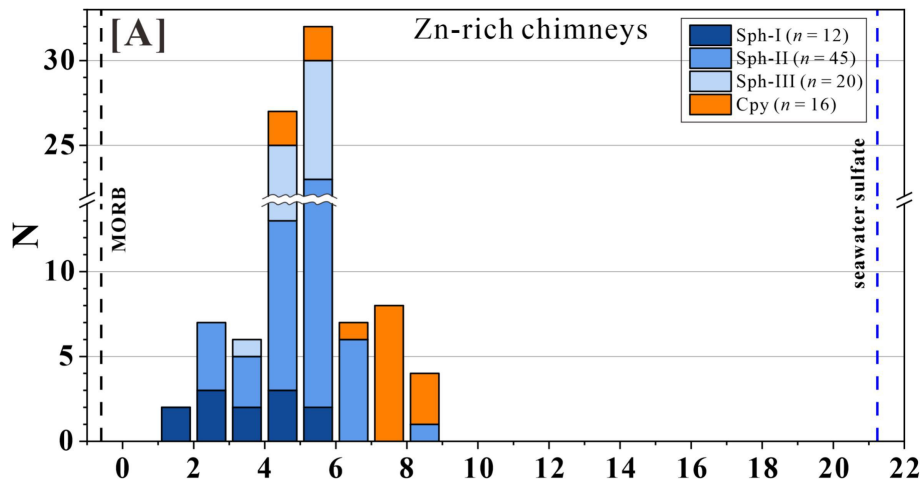


Figure 8